

Free Partially Commutative Structures

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In this paper, we study algebraic structures defined by a presentation of the form $\langle A; ab = ba \text{ for } (a, b) \in \mathcal{G} \rangle$. We show that these relators are the only one that can be interpreted as both monoid and Lie relators. We also study the free partially commutative monoids, Lie algebras, and groups: in all of these cases, we show how to obtain decomposition results for these structures into the corresponding free structures. © 1993 Academic Press, Inc.

INTRODUCTION

The *free partially commutative monoid* $M(A, \mathcal{G})$ was introduced by Cartier and Foata in 1969 in order to study combinatorial problems related to words rearrangements (cf. [CaFo]). By definition, $M(A, \mathcal{G})$ is the quotient of the free monoid A^* by a family of relators of the form

$$ab = ba \quad \text{for every } (a, b) \in \mathcal{G} \quad (\mathcal{P}ar\mathcal{C}om).$$

These relators are very particular since they can be interpreted as Lie relators ($[a, b] = 0$), as group commutators ($aba^{-1}b^{-1} = 1$), or as algebra relators ($ab - ba = 0$). In fact, as we see in this paper, they are the only monoid relators that also have a Lie interpretation. Therefore we can introduce the following algebraic structures, naturally related to partial commutations:

- the *free partially commutative group* $F(A, \mathcal{G})$ which is the quotient of the free group $F(A)$ by the normal subgroup generated by $\{aba^{-1}b^{-1}, (a, b) \in \mathcal{G}\}$.
- the *free partially commutative Lie algebra* $L_K(A, \mathcal{G})$ which is the quotient of the free Lie algebra $L_K(A)$ by the Lie ideal $\langle [a, b], (a, b) \in \mathcal{G} \rangle$.
- the *free partially commutative algebra* $K\langle A, \mathcal{G} \rangle$ which is the quotient of the free associative algebra $K\langle A \rangle$ by the ideal $\langle ab - ba, (a, b) \in \mathcal{G} \rangle$.

The study of partially commutative structures can be divided onto two main currents: first, there is a combinatorial current that is especially interested in studying the properties of the monoid $M(A, \mathfrak{g})$ and the extensions of the usual language theory to $M(A, \mathfrak{g})$ (cf. [Ch, Db, Mz, Me, Oc, Pe], for instance). In this area, there are also some group combinatorial studies of $F(A, \mathfrak{g})$ (see [Db], for instance). On the other hand, there is an algebraic viewpoint which takes a special interest in the study of $K\langle A, \mathfrak{g} \rangle$ or $L_K(A, \mathfrak{g})$ (see [Dc, Dr, KMNr, DK.1], for instance).¹ In this second direction, let us recall the two following interesting results:

— The free partially commutative associative algebra can be equipped with Hopf algebra structures (see [Sm]).

— The previous free partially structures are isomorphic iff the associated commutation graphs are isomorphic. More precisely, we have:

THEOREM 1. [KLMNR, Dr]. *Let K be a commutative domain and let A, B be two alphabets respectively equipped with the commutation relations $\mathfrak{g}_A$ and $\mathfrak{g}_B$. Then the following assertions are equivalent:*²

- (i) *The commutation graphs of $\mathfrak{g}_A$ and $\mathfrak{g}_B$ are isomorphic.*
- (ii) *The K -algebras $K\langle A, \mathfrak{g}_A \rangle$ and $K\langle B, \mathfrak{g}_B \rangle$ are isomorphic.*
- (iii) *The Lie algebras $L_K(A, \mathfrak{g}_A)$ and $L_K(B, \mathfrak{g}_B)$ are isomorphic.*
- (iv) *The groups $F(A, \mathfrak{g}_A)$ and $F(B, \mathfrak{g}_B)$ are isomorphic.*
- (v) *The monoids $M(A, \mathfrak{g}_A)$ and $M(B, \mathfrak{g}_B)$ are isomorphic.*

Continuing the pioneer studies of Schützenberger and Viennot (see [Sc, Vi.1, 2]), we are here at the intersection of these two viewpoints: indeed, the philosophy of this paper is to show that combinatorial results concerning $M(A, \mathfrak{g})$ are of great importance for obtaining semidirect decomposition results for the corresponding Lie algebras or groups. This explains why one of the first main result we obtain deals with $M(A, \mathfrak{g})$. More precisely, we will give a method that permits to construct free submonoids C_i^* such that we have:

THEOREM 2. *Let A be a finite alphabet equipped with a commutation*

¹ It should be noted that the expression “graph structures” has also been used in the literature to speak of partially commutative structures.

² Note that (v) $\Leftrightarrow$ (i) $\Rightarrow$ (iv) is obvious, that (iv) $\Rightarrow$ (iii) $\Rightarrow$ (ii) can easily be obtained by use of the Lie algebra constructed on the quotients of the lower central series of $F(A, \mathfrak{g}_A)$ (cf. [DK.2]) and by use of the enveloping algebra of $L_K(A, \mathfrak{g})$ (see also the remark ending [Dr]). Thus the only difficult part is (ii) $\Rightarrow$ (i) which is proved in [KMNr].

relation $\mathfrak{g}$. Then the free partially commutative monoid $M(A, \mathfrak{g})$ has a factorization into free monoids:

$$M(A, \mathfrak{g}) = (C_1)^* (C_2)^* \cdots (C_n)^*.$$

The codes $(C_i)_{i=1,n}$ which appear in the above factorization play an important role in all our results. It is also to be noted that the minimal number of free monoids that can occur (and that can be realized) in the above factorization is the *chromatic number* $\gamma(\mathfrak{g})$ of the commutation graph of $\mathfrak{g}$.

The previous theorem has a Lie algebra version which shows that every free partially commutative Lie algebra can be decomposed into a semidirect product of free Lie algebras as follows:

THEOREM 3. *Let A be a finite alphabet equipped with a commutation relation $\mathfrak{g}$ and let K be a commutative ring. Then the free partially commutative Lie algebra $L_K(A, \mathfrak{g})$ is isomorphic to a semidirect product of free Lie algebras:*

$$L_K(A, \mathfrak{g}) \simeq L_K(T_1) \ltimes (L_K(T_2) \ltimes (\dots L_K(T_n) \dots)).$$

There exists also a group version of the previous theorems which shows that every free partially commutative group has a semidirect decomposition into free groups as follows:

THEOREM 4. *Let A be a finite alphabet equipped with a commutation relation $\mathfrak{g}$. Then the free partially commutative group $F(A, \mathfrak{g})$ is isomorphic to a semidirect product of free groups:*

$$F(A, \mathfrak{g}) \simeq F(\mathcal{C}_1) \ltimes (F(\mathcal{C}_2) \ltimes (\dots F(\mathcal{C}_n) \dots)).$$

We must insist that the structure of the basic families of the free objects that appear in Theorems 3 and 4 is naturally related to the structure of the codes of Theorem 2. Indeed, every effective description of these codes immediately gives us an effective description of the basic families of Theorems 3 and 4. This explains why we study the monoid case closely.

Let us end this introduction with the structure of this paper. Section I deals with limitation results: we show here that the partial commutations are essentially the only monoid congruences that can be also interpreted as Lie relators. Section II is devoted to the study of $M(A, \mathfrak{g})$: we obtain here general methods of factorizations of $M(A, \mathfrak{g})$ into free monoids. Section III deals with the free partially commutative Lie algebra: it is here that we obtain a partially commutative version of Lazard's elimination theorem (see [Bo.5] or [La]). This result gives us immediately a semidirect

decomposition of $L(A, \mathfrak{g})$ into free Lie algebras. Finally, in Section IV, we give for $F(A, \mathfrak{g})$ the group versions of the Lie algebras results obtained in Section III.

0. PRELIMINARIES

(1) Free Monoids and Congruences

In the sequel, A^* denotes the *free monoid* generated by the alphabet A (see [Bo. 3, I, Sect. 7]). The elements of A^* are called *words* (see [Lo]). Let us also recall that a subset X of A^* is said to be a *code* iff the submonoid X^* generated by X is free (see [BePe] for more details).

Now let $w = a_1 \cdots a_n$ be a word in A^* . Then a word u is said to be a *subword* of w iff we have $u = a_{i_1} \cdots a_{i_k}$ for some $1 \leq i_1 < i_2 < \cdots < i_k \leq n$. Then the *complement* of the subword u in w is the subword of w denoted $\bar{u}$, defined by

$$\bar{u} = a_1 \cdots a_{i_1-1} a_{i_1+1} \cdots a_{i_2-1} a_{i_2+1} \cdots a_{i_k-1} a_{i_k+1} \cdots a_n.$$

We write $u \subseteq w$ when u is a subword of w and $u \subset w$ when u is a strict subword of w (i.e., a subword of w which is different from 1 and w).

Let A be an alphabet. Then we associate with every word w of A^* the three following notions of degree:

- the *partial degree* $|w|_a$ of w in $a \in A$ is just the number of a in w .
- the *total degree* $|w|$ of w is just the length of w .
- the *multidegree* of w is the A -uple $|w|_A = (|w|_a)_{a \in A} \in \mathbb{N}^{(A)}$.

Let us also recall that a *congruence* on a monoid M is an equivalence relation on M which is compatible with the laws of monoid (see [Ei.1, p. 9]). Let now A be an alphabet and let $\equiv$ be a congruence on A^* . Then we denote by $M(A; \equiv)$ the quotient monoid $A^*/\equiv$. Observe that every monoid can in fact be realized in this form (see [Bo.3, I, Sect. 7, No. 2]).

A congruence $\equiv$ on A^* is said to be *multihomogeneous* iff it satisfies the following property:

$$\forall u, v \in A^*, \quad u \equiv v \Rightarrow |u|_A = |v|_A.$$

Note. Multihomogeneous congruences seem to play a special role in the related algebraic structures. Note that the present paper can be considered as a contribution to the algebraic study of such congruences of degree 2.

(2) Alphabetic Congruences

Let A be an alphabet and let $\lambda, \mathfrak{g}$ be two symmetric subsets of $A \times A - \Delta_A$, where $\Delta_A = \{(a, a), a \in A\}$ is the diagonal of $A \times A$. Then the

alphabetic congruence over A associated with the pair (λ, ϑ) is the finest monoid congruence over A^* , denoted $\equiv_{\lambda, \vartheta}$, such that we have for all letters $a, b \in A$:

$$a \equiv_{\lambda, \vartheta} b \Leftrightarrow (a, b) \in \lambda \quad \text{and} \quad ab \equiv_{\lambda, \vartheta} ba \Leftrightarrow (a, b) \in \vartheta.$$

The associated monoid $M(A; \equiv_{\lambda, \vartheta})$ will be denoted by $M(A; \lambda, \vartheta)$ in the sequel.

Notes. (1) When $\lambda = \emptyset$, we denote by $\equiv_{\vartheta}$ the previous congruence, which is classically called a *partial commutation relation* on A . In this case, we will also denote by $M(A; \vartheta)$ the associated monoid $M(A; \equiv_{\vartheta})$, which is called the *free partially commutative monoid*.

(2) When $\vartheta = \emptyset$, we also denote by $\equiv_{\lambda}$ the previous congruence, which is here a *letter identification congruence*.

Remarks. (1) Let (λ, ϑ) be an alphabetic congruence on an alphabet A , let $[\lambda]$ be the equivalence relation on A generated by λ , and let π_{λ} be the canonical surjection from A onto $\mathcal{A} = A/[\lambda]$. Then, if we denote by $\vartheta_{\mathcal{A}}$ the subset of $\mathcal{A} \times \mathcal{A} - \Delta_{\mathcal{A}}$ which is defined by

$$\forall \alpha \neq \beta \in \mathcal{A}, (\alpha, \beta) \in \vartheta_{\mathcal{A}} \Leftrightarrow \exists (a, b) \in \vartheta, \pi_{\lambda}(a) = \alpha, \pi_{\lambda}(b) = \beta,$$

it is easy to prove that the monoid $M(A; \lambda, \vartheta)$ is isomorphic to the free partially commutative monoid $M(\mathcal{A}; \vartheta_{\mathcal{A}})$. Hence, each global property of the free partially commutative monoids holds also for $M(A; \lambda, \vartheta)$.

(2) Let (λ, ϑ) be an alphabetic congruence on an alphabet A and let w be an element of $M(A; \lambda, \vartheta)$. Then, we call *alphabet* of w the subset of A , denoted $\text{Alph}(w)$, which is defined by:

$$\text{Alph}(w) = \{a \in A, \exists u, v \in M(A; \lambda, \vartheta), w = u \cdot a \cdot v\}$$

We shall need the following definition that was classically given in the partial commutation case (see [Db, p. 31] for instance):

DEFINITION 0.1. Let (λ, ϑ) be an alphabetic congruence over an alphabet A and let x, y be two elements in $M(A; \lambda, \vartheta)$. Then x and y are said to be *independent* for (λ, ϑ) iff we have $a \not\equiv_{\lambda} b$ and $ab \equiv_{\lambda, \vartheta} ba$ for every letter $a \in \text{Alph}(x)$ and $b \in \text{Alph}(y)$.

The following result, which is a generalization of Levi's lemma, was proved in the partially commutative case in [CP] (see also [Db, Proposition 3.1.1]). Hence, by a previous remark, it holds also for alphabetic congruences that

PROPOSITION 0.1. *Let (λ, ϑ) be an alphabetic congruence over an alphabet A and let x, y, z, t be elements in $M(A; \lambda, \vartheta)$ such that $xy = zt$. Then there exist $u, v, r, s \in M(A; \lambda, \vartheta)$, where r, s are independent for (λ, ϑ) , such that*

$$x = ur, \quad y = sv, \quad z = us, \quad t = rv.$$

(3) Partial Commutations

We present here some definitions concerning especially partial commutation relations. Note that this framework was worked out in depth by many people [Db, CP, Mz, Me, Oc, Zi]. Let us now consider an alphabet A equipped with a partial commutation relation ϑ . One represents ϑ classically by its *commutation graph* whose vertices are the letters of A and whose edges are the pairs of letters (a, b) of $A \times A$ such that $(a, b) \in \vartheta$ (cf. [Ch] for more details). Sometimes it is also useful to consider the *non-commutation graph* $\bar{\vartheta}$ associated with ϑ : its vertices are the letters of A and an edge exists between two letters $a, b \in A$ iff $(a, b) \notin \vartheta$. Finally let us also recall that $M(A, \vartheta)$ is cancellative (see [Ch, Corollary I.2.5 or Proposition IV.1 below, which claims that $M(A, \vartheta)$ embeds in fact in a group).

DEFINITION 0.2. Let A an alphabet equipped with a commutation relation ϑ . Then a subalphabet B of A is said to be *non-commutative* for ϑ iff we have

$$\forall a, b \in B, \quad (a, b) \notin \vartheta.$$

Note. According to the classical terminology in graph theory, a subalphabet B of A is non-commutative if and only if B is *stable* in the commutation graph of ϑ (see [Bg, Chap. 13] for instance).

(4) Factorizations of a Monoid

Let M be a monoid and let I be a totally ordered set. Then a family $(M_i)_{i \in I}$ of submonoids of M is said to be a *factorization* of M iff every element m of M has a unique decomposition of the form

$$m = m_{i(1)} \cdots m_{i(n)},$$

where $i(1) < \cdots < i(n)$ and $m_{i(j)} \in M_{i(j)}$ for every $j \in \llbracket 1, n \rrbracket$. We denote such a situation as follows:

$$\underline{M} = \prod_{i \in I} \underline{M}_i. \quad (\mathcal{F})$$

Remark. If we denote for a subset $N \subset M$ by $\underline{N}$ the characteristic series of N , i.e., the series of $\mathbb{N}\langle\langle M \rangle\rangle$ defined by

$$(\underline{N}|m) = 0 \text{ when } m \notin N \quad \text{and} \quad (\underline{N}|m) = 1 \text{ when } m \in N$$

(see also [BePe, SaSo]), relation $(\mathcal{F})$ can be interpreted as an equality in $\mathbb{N}\langle\langle M \rangle\rangle$. This motivates our above choice for denoting a factorization.

The following result shows how factorizations can be composed. Its proof is easy and is left to the reader.

PROPOSITION 0.2. *Let $(J(i))$ be a family of totally ordered sets, indexed by a totally ordered set I , let S denote its ordinal sum (cf. [Bo.1, III, 69]), let $(M_i)_{i \in I}$ be a factorization of M , and let $(M_j^i)_{j \in J(i)}$ be a factorization of M_i for every $i \in I$. Then the family $(M_j^i)_{(i,j) \in S}$ is a factorization of M .*

(5) Algebras and Lie Algebras

In all the sequel, K will denote a non-trivial commutative ring.

Let A be an alphabet. Then $K\langle A \rangle$ will denote the free associative K -algebra over A , i.e., the K -algebra of the non-commutative polynomials over A . The homogeneous components of $K\langle A \rangle$ for the multidegree are defined by

$$K_\alpha\langle A \rangle = \bigoplus_{|w|_A = \alpha} K \cdot w$$

for every multidegree $\alpha \in \mathbb{N}^{(A)}$. Now let I be a two-sided ideal of $K\langle A \rangle$: then we denote by $K\langle A; I \rangle$ the quotient K -algebra $K\langle A \rangle/I$.

The reader is referred to [Bo.4] or to [Ja] for the usual definitions and properties concerning Lie K -algebras. Let us now recall some basic facts about derivations:

DEFINITION 0.3. Let $\mathcal{G}$ be a Lie K -algebra. Then a K -module endomorphism D of $\mathcal{G}$ is said to be a *derivation* of $\mathcal{G}$ iff we have

$$D([x, y]) = [Dx, y] + [x, Dy] \quad \text{for every } x, y \in \mathcal{G}.$$

Remark. The following facts are classical (cf. [Bo.4] or [Ja]):

- (i) For every $x \in \mathcal{G}$, the mapping $\text{ad } x: y \rightarrow [x, y]$ is a derivation.
- (ii) The set $\text{Der}(\mathcal{G})$ of the derivations of $\mathcal{G}$ is a Lie subalgebra of $\text{End}_K(\mathcal{G})$.
- (iii) If a derivation is 0 on a generating set of $\mathcal{G}$, it is 0 everywhere.

DEFINITION 0.4. Let $\mathcal{G}, \mathcal{H}$ be two Lie algebras and let φ be a Lie

morphism from $\mathcal{H}$ into $\text{Der}(\mathcal{G})$. Then the *semidirect product* of $\mathcal{G}$ and $\mathcal{H}$ relative to φ is the Lie algebra denoted $\mathcal{G} \ltimes_{\varphi} \mathcal{H}$ build over the set $\mathcal{G} \times \mathcal{H}$ whose law is defined by

$$[(g, h), (g', h')] = ([g, g'] + \varphi(h)(g') - \varphi(h')(g), [h, h'])$$

for every g, g' in $\mathcal{G}$ and h, h' in $\mathcal{H}$.

Let us now recall some properties of the free Lie algebra (see [Bo.5] or [Vi.2] for more details):

DEFINITION 0.5. Let A be an alphabet. Then we call *free Lie algebra* over A and denote by $L_K(A)$, or $L(A)$ if there is no ambiguity, the Lie subalgebra of $K\langle A \rangle$ that is generated by A .

Notes. (1) The free Lie algebra $L(A)$ is in fact the object freely generated by A in the Lie algebra category. It solves the associated universal problem of factorizing any mapping from A into a Lie algebra $\mathcal{G}$ (see [Lo, Corollary 5.3.9]).

(2) Let δ be a mapping from A into $L(A)$. Then δ can be extended in a unique way as a derivation of $L(A)$.

(3) For every word $w = a_1 \cdots a_n \in A^*$, we denote by $\text{ad } w$ the K -module endomorphism of $L(A)$ defined by

$$\text{ad } w = \text{ad } a_1 \circ \cdots \circ \text{ad } a_n.$$

I. LIE THEORY VERSUS MONOID THEORY

In this section, we show that the partially commutative theory is in a certain sense the only one that permits a Lie theory and a monoid theory to be developed together.

(1) *Monoidal Ideals of $K\langle A \rangle$*

We will study here some properties of the two-sided ideals of $K\langle A \rangle$ that are generated by a monoid congruence of A^* , i.e. that are defined as follows:

DEFINITION I.1. Let I be a two-sided ideal of $K\langle A \rangle$. Then we say that I is a *monoidal ideal* iff there exists a congruence $\equiv$ on A^* such that

$$I = I_{\equiv} = \sum_{u \equiv v} K\langle A \rangle \cdot (u - v) \cdot K\langle A \rangle \left(= \sum_{u \equiv v} K \cdot (u - v) \right).$$

Notations. If $\equiv$ is a congruence over A^* , we denote by $K\langle A; \equiv \rangle$ the quotient K -algebra $K\langle A; I_{\equiv} \rangle = K\langle A \rangle / I_{\equiv}$.

Note. It can be easily shown that the congruence $\equiv$ is uniquely determined by the monoidal ideal $I_{\equiv}$ (see Lemma I.3 below).

The following proposition shows that the K -algebra $K\langle A; \equiv \rangle$ can be also seen as the algebra of the monoid $M(A; \equiv)$:

PROPOSITION I.1. *Let $\equiv$ be a congruence on A^* . Then the K -algebras $K\langle A \rangle / I_{\equiv}$ and $K[M(A; \equiv)]$ are canonically isomorphic.*

Proof. This result can easily be proved with the same methods that were used in [DK.1] or in [Du] for the partially commutative case. ■

The next result characterizes the graded monoidal ideals of $K\langle A \rangle$ for the multidegree (cf. [Bo.2] for more details concerning gradations):

PROPOSITION I.2. *Let $I = I_{\equiv}$ be a monoidal two-sided ideal of $K\langle A \rangle$. Then the two following statements are equivalent:*

- (i) *I is graded for the multidegree.*
- (ii) *The congruence $\equiv$ is multihomogeneous.*

Proof. The implication (ii) $\Rightarrow$ (i) is obvious. Therefore we need only prove that (i) $\Rightarrow$ (ii). Thus let us consider $I = I_{\equiv}$, a two-sided monoidal ideal of $K\langle A \rangle$ graded for the multidegree and associated with a congruence $\equiv$ of A^* . This means exactly that we have the relation

$$I = \bigoplus_{\alpha \in \mathbb{N}(A)} (I \cap K_{\alpha}\langle A \rangle). \quad (\mathcal{G}, \mathcal{M})$$

Let us suppose that (ii) were false: then there would exist for instance two words $w \equiv w'$ with $w \in K_{\alpha}\langle A \rangle$, $w' \in K_{\beta}\langle A \rangle$, and $\alpha \neq \beta$. But, by definition of a monoidal ideal, we have $w - w' \in I$. It follows according to $(\mathcal{G}, \mathcal{M})$ that

$$w - w' = \sum_{\mu \in \mathbb{N}(A)} i_{\mu},$$

where i_{μ} belongs to $I \cap K_{\mu}\langle A \rangle$ for every μ . Then, since $\alpha \neq \beta$, if we read this relation on the multidegree graduation $(K_{\alpha}\langle A \rangle)_{\alpha \in \mathbb{N}(A)}$, we easily obtain

$$w \in I \cap K_{\alpha}\langle A \rangle \quad \text{and} \quad w' \in I \cap K_{\beta}\langle A \rangle. \quad (\mathcal{A}, \mathcal{B})$$

Let us now consider the graph $G = (A^*, E)$, where two vertices u, v in A^* are related by an edge iff $u \equiv v$. Let $(C_r)_{r \in R}$ denote the connected

components of G and let us choose a vertex u_r in each component C_r in such a way that $w = u_{r_1}$ with $r_1 \in R$. Now for every r in R , let

$$C_r^- = C_r - \{u_r\}.$$

Then we can give the following lemma which makes precise the structure of I . Its proof is easy and is left to the reader.

LEMMA 1.3. *We have the direct sum decomposition*

$$I = \bigoplus_{r \in R} \bigoplus_{u \in C_r^-} K \cdot (u_r - u).$$

Using this lemma and $(\mathcal{AB})$, it is easy to see that we can write

$$\begin{aligned} w = u_{r_1} &= \sum_{u \in C_{r_1}^-} k_{r_1}^u (u_{r_1} - u) \\ &\Leftrightarrow \left(1 - \sum_{u \in C_{r_1}^-} k_{r_1}^u\right) u_{r_1} + \sum_{u \in C_{r_1}^-} k_{r_1}^u u = 0, \end{aligned}$$

where the elements $k_{r_1}^u$ belong to K . It is now straightforward to conclude from this last identity that every $k_{r_1}^u$ is 0. Therefore u_{r_1} is equal to 0 in $K\langle A \rangle$. Hence we have obtained a contradiction, which permits us to conclude our proof. ■

(7) Bialgebras and Alphabetic Commutations

We refer to [Bo.3, Bo.5] for all basic properties concerning coalgebras and bialgebras. We can now give the main theorem of this section which characterizes the alphabetic congruences in terms of bialgebras:

THEOREM 1.4. *Let K be a ring of characteristic 0, let A be an alphabet, let $\equiv$ be a congruence on A^* , and let $I_\equiv$ be the corresponding two-sided monoidal ideal of the free associative algebra $K\langle A \rangle$. Then the two following statements are equivalent:*

(i) *The K -algebra $K\langle A; I \rangle$ can be equipped with a bialgebra structure where the letters are primitive for the comultiplication c ; i.e.,*

$$\forall a \in A, \quad c(a) = 1 \otimes a + a \otimes 1. \quad (\text{Prim})$$

(ii) *The congruence $\equiv$ is an alphabetic congruence on A^* .*

Proof. The implication (ii) $\Rightarrow$ (i) is easy and comes from [Sm]. Thus we just have to prove that (i) $\Rightarrow$ (ii). Thus let us denote by c a comultiplication on $K\langle A; I \rangle$ which satisfies the relations (Prim). Let us also denote by

$[w]$ the class of a word $w \in A^*$ in $M(A; \equiv)$. Thus it is easy to see that we have in the K -algebra $K\langle A; I \rangle \simeq K[M(A; \equiv)]$ (by Proposition I.1):

$$\forall w \in A^*, \quad c([w]) = \sum_{u \subseteq w} [u] \otimes [\bar{u}]. \quad (1)$$

Therefore the following relation holds for every pair of words $w \equiv w'$:

$$\sum_{u \subseteq w} [u] \otimes [\bar{u}] = \sum_{u \subseteq w'} [u] \otimes [\bar{u}]. \quad (2)$$

It follows easily that we have in fact for every pair of words $w \equiv w'$

$$\sum_{u \subset w} [u] \otimes [\bar{u}] = \sum_{u \subset w'} [u] \otimes [\bar{u}]. \quad (3)$$

Since $K\langle A; I \rangle \simeq K[M(A; \equiv)]$, Corollary 2 of [Bo.3, III, Sect. 3, No. 7] implies also easily that the following family is a basis of the free K -module $K\langle A; I \rangle \otimes K\langle A; I \rangle$:

$$(m \otimes n)_{(m, n) \in M(A; \equiv) \times M(A; \equiv)}. \quad (\mathcal{B})$$

These preliminaries being given, we can now prove that the congruence $\equiv$ is necessarily homogeneous for the total degree:

LEMMA I.5. *Under the previous hypotheses, we have*

$$\forall w, w' \in A^*, \quad w \equiv w' \Rightarrow |w| = |w'|.$$

Proof. Let $w \equiv w'$ be two words of A^* equivalent for $\equiv$. By (2), we have

$$\sum_{u \subseteq w} [u] \otimes [\bar{u}] = \sum_{u \subseteq w'} [u] \otimes [\bar{u}].$$

If we decompose these two elements of $K\langle A; I \rangle \otimes K\langle A; I \rangle$ on the K -basis $(\mathcal{B})$, we obtain a relation in $K\langle A; I \rangle \otimes K\langle A; I \rangle$ of the form

$$\sum_{m, n \in M(A; \equiv)} \lambda_{m, n} m \otimes n = \sum_{m, n \in M(A; \equiv)} \mu_{m, n} m \otimes n,$$

where $\lambda_{m, n}$ and $\mu_{m, n}$ are elements of $\mathbb{Z} \subset K$, since K is of characteristic 0. As the previous decomposition holds on a basis of $K\langle A; I \rangle$, we easily obtain

$$\sum_{m, n \in M(A; \equiv)} \lambda_{m, n} = \sum_{m, n \in M(A; \equiv)} \mu_{m, n}.$$

But these two sums are clearly respectively equal to the numbers of subwords of w and of w' . Thus we have

$$2^{|w|} = 2^{|w'|} \Rightarrow |w| = |w'|.$$

Therefore the lemma is proved. ■

We now prove that $\equiv$ is necessarily cancellative. To this end, let us first show that, when K is a field, the bialgebra $K\langle A; I \rangle$ is the enveloping bialgebra of the Lie algebra of its primitive elements:

LEMMA I.6. *Let K be a field of characteristic 0. Let us denote by $P(A; I)$ the Lie algebra of the primitive elements of $K\langle A; I \rangle$. Then the enveloping algebra $U(P(A; I))$ is isomorphic to $K\langle A; I \rangle$.*

Proof. Observe first that the bialgebra $(K\langle A; I \rangle, c)$ is clearly cocommutative. Let us consider the family $(E_n)_{n \geq 0}$ defined by

$$E_n = \sum_{|w| \leq n} K \cdot [w].$$

It is now clear that this family is a filtration on $K\langle A; I \rangle$ that is compatible with the bialgebra structure. Our lemma follows now immediately from Theorem 1 of [Bo.5, II, Sect. 1, No. 6]. ■

We can now show that $\equiv$ is left and right cancellative:

LEMMA I.7. *Let u, v, w be words in A^* . Then we have*

$$uw \equiv vw \Rightarrow u \equiv v \quad \text{and} \quad wu \equiv wv \Rightarrow u \equiv v.$$

Proof. As the arguments are similar, we can restrict ourselves to proving the first implication. At first, observe that $M(A; \equiv)$ embeds into $\mathbb{Z}\langle A; \equiv \rangle$ according to Proposition I.1. But it follows from the previous lemma that the algebra $\mathbb{Q}\langle A; \equiv \rangle$ is an enveloping algebra and hence a domain (cf. Corollary 7 of [Bo.4, I, Sect. 2, No. 7]). Therefore, since we have $\mathbb{Z}\langle A; \equiv \rangle \subset \mathbb{Q}\langle A; \equiv \rangle$, it follows that $\mathbb{Z}\langle A; \equiv \rangle$ is itself a domain. Thus if the first relation holds, we can deduce that $u = v$ in $\mathbb{Z}\langle A; \equiv \rangle$. Since by Proposition I.1, $\mathbb{Z}\langle A; \equiv \rangle \simeq \mathbb{Z}[M(A; \equiv)]$, we conclude that $u \equiv v$. ■

Let us denote by λ and ϑ the following families of $A \times A$:

$$\lambda = \{(a, b) \in A \times A - \Delta_A, a \equiv b\}$$

and

$$\vartheta = \{(a, b) \in A \times A - \Delta_A, ab \equiv ba\}.$$

We now prove by induction on n that

$$\forall w, w' \in A^*, \quad |w| = |w'| = n, w \equiv w' \Rightarrow w \equiv_{\lambda, g} w'.$$

Note that the converse result is obvious and that we make use of it in the sequel. Therefore this result immediately implies that the congruence $\equiv$ is an alphabetic congruence according to Lemma I.5. But let us first show that it holds for $n = 2$:

LEMMA I.8. *Let w, w' be two words of length 2 that are equivalent by $\equiv$. Then w is equivalent to w' by the alphabetic congruence $\equiv_{\lambda, g}$.*

Proof. Let us suppose that $w = ab$ and $w' = cd$, where a, b, c, d are four letters of A . Thus we have $ab \equiv cd$ ($\mathcal{E}$). According to (3), we have

$$[a] \otimes [b] + [b] \otimes [a] = [c] \otimes [d] + [d] \otimes [c].$$

Since the ring K is of characteristic 0, it follows that

$$[a] \otimes [b] = [c] \otimes [d] \quad \text{or} \quad [a] \otimes [b] = [d] \otimes [c].$$

If the first relation holds, it follows that $a \equiv c$ and $b \equiv d$. In this case, we clearly have $ab \equiv_{\lambda} cd$. If the second relation holds, it follows that $a \equiv d$ and $b \equiv c$. Thus, it comes from ($\mathcal{E}$) that $ab \equiv ba$. Hence, it follows easily that we have $ab \equiv_{\lambda, g} cd$ in every case. This ends our proof. ■

Let us suppose that our induction hypothesis holds at every order $n \geq 2$. Then let w, w' be two words of length $n + 1$ which are equivalent by $\equiv$. We can write $w = au$, with $a \in A$ and $|u| = n$. Since we are in characteristic 0, it follows from (3) and Lemma I.5 that there exists a letter b of w' such that

$$[a] \otimes [u] = [b] \otimes [\bar{b}] \Rightarrow a \equiv b \text{ and } u \equiv \bar{b}.$$

If we write $w' = u_1 bu_2$, we have, by the induction hypothesis,

$$a \equiv_{\lambda} b \quad \text{and} \quad u \equiv_{\lambda, g} u_1 u_2.$$

It follows immediately from Lemma I.7 that we have

$$bu_1 u_2 \equiv u_1 bu_2 \Rightarrow bu_1 \equiv u_1 b.$$

If $|u_2| > 1$, we can apply the induction hypothesis to obtain

$$bu_1 \equiv_{\lambda, g} u_1 b \Rightarrow w = au \equiv_{\lambda, g} bu_1 u_2 \equiv_{\lambda, g} u_1 bu_2 = w'.$$

Thus the proof ends in this case. In the other case, $u_2 = 1$ and it is easy to see that the problem can be reduced to prove the following lemma:

LEMMA I.9. *Let $a_0, a_1, \dots, a_n$ be letters of A . Then we have*

$$a_0 a_1 \cdots a_n \equiv a_1 \cdots a_n a_0 \Rightarrow a_0 a_1 \cdots a_n \equiv_{\lambda, \vartheta} a_1 \cdots a_n a_0. \quad (\mathcal{R})$$

Proof. The first above assumption implies that

$$c(a_0 a_1 \cdots a_n) = c(a_1 \cdots a_n a_0).$$

Developing this identity according to (3), it is easy to prove that there exists an index $i \in \llbracket 0, n \rrbracket$ such that

$$[a_i] \otimes [a_0 \cdots a_{i-1} a_{i+1} \cdots a_n] = [a_n] \otimes [a_1 \cdots a_{n-1} a_0]. \quad (4)$$

If $i = 0$, it follows immediately that: $a_0 \equiv a_n$. Therefore, we have

$$a_0 a_1 \cdots a_n \equiv a_1 \cdots a_n a_n \Rightarrow a_0 \cdots a_{n-1} \equiv a_1 \cdots a_n$$

by Lemma I.7. Then, since $a_0 \equiv_{\lambda} a_n$, it follows easily by application of the induction hypothesis that the second member of $(\mathcal{R})$ holds.

Let us suppose now that $i \neq 0$. Then it follows from relation (4) and from the induction hypothesis that

$$a_i \equiv_{\lambda} a_n \quad \text{and} \quad a_0 \cdots a_{i-1} a_{i+1} \cdots a_n \equiv_{\lambda, \vartheta} a_1 \cdots a_{n-1} a_0.$$

By application of Proposition 0.1, there exists four words u, r, s, v in A^* with r and s independent for (λ, ϑ) such that

$$\begin{aligned} a_0 &\equiv_{\lambda, \vartheta} ur, & a_1 \cdots a_{i-1} a_{i+1} \cdots a_n &\equiv_{\lambda, \vartheta} sv \\ a_1 \cdots a_{n-1} &\equiv_{\lambda, \vartheta} us, & a_0 &\equiv_{\lambda, \vartheta} rv. \end{aligned}$$

We must now consider two different cases in order to conclude.

(a) $r = 1$. If $r = 1$, we have $a_0 \equiv_{\lambda} v \equiv_{\lambda} u$. It follows that

$$a_1 \cdots a_{n-1} \equiv_{\lambda, \vartheta} a_0 s \Rightarrow a_1 \cdots a_{n-1} \equiv a_0 s.$$

But the first statement of $(\mathcal{R})$ implies that

$$a_0 a_1 \cdots a_n \equiv a_1 \cdots a_n a_0 \Rightarrow a_0 a_1 \cdots a_n \equiv a_0 s a_n a_0.$$

Therefore we deduce by Lemma I.7 and by the induction hypothesis that

$$a_1 \cdots a_n \equiv s a_n a_0 \Rightarrow a_1 \cdots a_n \equiv_{\lambda, \vartheta} s a_n a_0.$$

Thus we easily obtain

$$a_0 a_1 \cdots a_n \equiv_{\lambda, \vartheta} a_0 s a_n a_0 \equiv_{\lambda, \vartheta} a_1 \cdots a_{n-1} a_n a_0.$$

(b) $r \neq 1$. In this case, we necessarily have $u = v = 1$ and $r \equiv_{\lambda} a_0$. Thus, it follows from previous relations that we have

$$a_1 \cdots a_{n-1} \equiv_{\lambda, \vartheta} s.$$

Moreover r, s are independent for (λ, ϑ) . Hence a_0 and s are also independent for (λ, ϑ) . Therefore $a_0 a_1 \equiv_{\lambda, \vartheta} a_1 a_0$ ($\mathcal{C}_{01}$). Thus applying the first statement of ($\mathcal{R}$), we obtain the relation

$$a_0 a_1 a_2 \cdots a_n \equiv a_1 a_2 \cdots a_n a_0 \Rightarrow a_1 a_0 a_2 \cdots a_n \equiv a_1 \cdots a_n a_0,$$

which gives us immediately by Lemma I.7 and by the induction hypothesis

$$a_0 a_2 \cdots a_n \equiv_{\lambda, \vartheta} a_2 \cdots a_n a_0.$$

Then, multiplying by a_1 on the left of this identity, we obtain

$$a_1 a_0 a_2 \cdots a_n \equiv_{\lambda, \vartheta} a_1 a_2 \cdots a_n a_0 \Rightarrow a_0 a_1 a_2 \cdots a_n \equiv_{\lambda, \vartheta} a_1 a_2 \cdots a_n a_0$$

by using ($\mathcal{C}_{01}$). This ends the proof of our lemma. ■

Thus our induction hypothesis is proved at order $n+1$. Hence we showed that the congruence $\equiv$ is equal to $\equiv_{\lambda, \vartheta}$ and our proof is now complete. ■

Notes. (1) When K is of finite characteristic $p > 0$, the same result does not hold. Indeed, let us consider the congruence $\equiv$ defined by $a \equiv a^p$ over a one letter alphabet $\{a\}$. Then let us introduce the $\mathbb{F}_p$ -algebra $\mathbb{F}_p \langle a; \equiv \rangle$. It is easy to see that we can equip it with a bialgebra structure by defining

$$c \left(\sum_{i=0}^{p-1} k_i a^i \right) = \sum_{i=0}^{p-1} k_i (1 \otimes a + a \otimes 1)^i.$$

Hence $\mathbb{F}_p \langle A; \equiv \rangle$ is a counterexample to the previous theorem in this case.

(2) It is an open problem how to associate a "good" Lie $\mathbb{Z}$ -algebra to an arbitrary congruence $\equiv$ over A^* . Note that the proof of the previous theorem seems to indicate that the congruence $\equiv$ must be cancellative if some real work is to be done.

COROLLARY I.10. *Let K be a ring of characteristic 0, let A be an alphabet, let $\equiv$ be a congruence on A^* , and let $I_{\equiv}$ be the associated two-sided monoidal ideal of the free associative algebra $K \langle A \rangle$. Then the two following statements are equivalent:*

- (i) The K -algebra $K\langle A; I \rangle$ is the enveloping algebra of a Lie K -algebra $\mathcal{G}$ in such a way that the alphabet A is contained in $\sigma(\mathcal{G})$.³
- (ii) The congruence $\equiv$ is an alphabetic congruence on A^* .

Proof. The implication (ii) $\Rightarrow$ (i) is easy (cf. [DK.1]). Let us suppose now that (i) holds. Thus there exists a Lie K -algebra $\mathcal{G}$ such that $K\langle A; I \rangle \simeq U(\mathcal{G})$ and $A \subset \sigma(\mathcal{G})$. But, by Proposition 7 of [Bo.5, II, Sect. 1, No. 4], there exists on $U(\mathcal{G})$ a unique bialgebra structure such that every element of $\sigma(\mathcal{G})$ is primitive. Then if we denote by c the corresponding comultiplication, it follows that

$$\forall a \in A, \quad c(a) = 1 \otimes a + a \otimes 1.$$

Therefore we can apply Theorem I.4 in order to obtain our result. ■

Note. This result shows the difficulty of trying to associate a general congruence on A^* with a "good" Lie $\mathbb{Z}$ -algebra.

COROLLARY I.11. *Let K be a ring of characteristic 0, let A be an alphabet, let $\equiv$ be a congruence on A^* , and let $I_{\equiv}$ be the classical two-sided monoidal ideal of $K\langle A \rangle$. Let us denote by $\mathcal{L}\langle A; \equiv \rangle$ the Lie subalgebra of $K\langle A; I \rangle$ generated by the letters. Then the following statements are equivalent:*

- (i) $K\langle A; \equiv \rangle$ is the enveloping algebra of $\mathcal{L}\langle A; \equiv \rangle$.
- (ii) The congruence $\equiv$ is an alphabetic congruence on A^* .

Proof. The implication (ii) $\Rightarrow$ (i) can be found in [DK.1]. Conversely the other implication follows from the previous corollary. ■

Note. Again this result does not hold when $\text{char } K = p > 0$. Indeed, let us consider the finest congruence $\equiv$ on $\{a, b\}^*$ such that $a^p b \equiv b a^p$. Observe that this relation can here be written $\text{ad}^p(a) \cdot b = 0$; hence it follows easily that we have $\mathcal{U}(\mathcal{L}\langle A; \equiv \rangle) = K\langle A; \equiv \rangle$ for this congruence.

(3) Monoidal and Lie Type Ideals of $K\langle A \rangle$

In this section, we study the connections between the Lie theory and the monoid theory from the viewpoint of the two-sided ideals of $K\langle A \rangle$.

DEFINITION I.2. Let I be a two-sided ideal of the free associative K -algebra $K\langle A \rangle$ over A . Then we say that I is of *Lie type* iff there exists a Lie ideal J of $L(A)$ such that

$$I = I(J) = J \cdot K\langle A \rangle (= K\langle A \rangle \cdot J = K\langle A \rangle \cdot J \cdot K\langle A \rangle).$$

³ σ denotes the canonical morphism from $\mathcal{G}$ in $U(\mathcal{G})$ (see [Bo.4]).

Then we have the following theorem that follows from the main result of the previous subsection:

THEOREM I.12. *Let K be a ring of characteristic 0, let A be an alphabet, and let I be a two-sided ideal of the free associative algebra $K\langle A \rangle$. Then the two following statements are equivalent:*

- (i) *I is a monoidal ideal of Lie type.*
- (ii) *There exists an alphabetic congruence (λ, ϑ) on A^* such that*

$$I = I \equiv_{\lambda, \vartheta}$$

Proof. Clearly, the only thing to prove is that (i) $\Rightarrow$ (ii). Let us consider now a two-sided ideal I of $K\langle A \rangle$ which is monoidal and of Lie type. Since I is of Lie type, there exists a Lie ideal J of $L(A)$ such that $I = K\langle A \rangle \cdot J \cdot K\langle A \rangle$. Hence we can consider the Lie K -algebra $\mathcal{G} = L(A)/J$. According to Proposition 3 of [Bo.4, I, Sect. 2, No. 3] the enveloping algebra of $\mathcal{G}$ is given by

$$U(\mathcal{G}) \simeq U(L(A))/\langle \sigma(J) \rangle \simeq K\langle A \rangle / I = K\langle A; I \rangle, \quad (1)$$

where σ denotes the canonical morphism of $L\langle A \rangle$ into its enveloping algebra $K\langle A \rangle$ which here is into (see Theorem 1 of [Bo.5, II, Sect. 3, No. 1]). Hence it suffices now to apply Corollary I.10 to conclude. ■

COROLLARY I.13. *Let K be a ring of characteristic 0, let A be an alphabet, and let I be a two-sided ideal of the free associative algebra $K\langle A \rangle$. Then the two following statements are equivalent:*

- (i) *I is a monoidal ideal of Lie type, graded for the multidegree.*
- (ii) *There exists a partial commutation ϑ on A^* such that*

$$I = I \equiv_{\vartheta}$$

Proof. The implication (ii) $\Rightarrow$ (i) is obvious. Conversely, (i) $\Rightarrow$ (ii) follows easily from Proposition I.2 and Theorem I.12. ■

(4) Subword Monoids

We shall now study the monoids for which a subword/complement notion is possible to define. First let us make this concept precise:

DEFINITION I.3. Let M be a monoid. Then M is said to be a *subword monoid* if and only if there exist

(1) a mapping σ (the *subword mapping*) from M into $\mathcal{P}(M)$ such that we have, for every m in M , $\{1, m\} \subset \sigma(m)$ and, for every m, n in M , $\sigma(m) \cdot \sigma(n) \subset \sigma(mn)$, and such that the following mapping is a bijection:

$$\sigma(m) \times \sigma(n) \rightarrow \sigma(mn)$$

$$(u, v) \rightarrow uv;$$

(2) for every $m \in M$, a mapping χ_m (the *complement mapping* relative to m) from $\sigma(m)$ into M which satisfies, for every $m, n \in N$,

$$\forall u \in \sigma(m), \forall v \in \sigma(n), \quad \chi_m(u) \chi_n(v) = \chi_{mn}(uv).$$

PROPOSITION I.14. *Let $M = M(A; \equiv)$ be a presented monoid. Then the following statements are equivalent:*

- (i) *M is a subword monoid such that $\sigma(a) = \{1, a\}$ for every $a \in A$.*
- (ii) *M is a free partially commutative monoid.*

Proof. The implication (ii) $\Rightarrow$ (i) comes from [Sm]. Conversely, let M be a subword monoid. Then we can equip the $\mathbb{Z}$ -algebra $\mathbb{Z}[M]$ with a comultiplication c defined by $\mathbb{Z}$ -linearity from the relations

$$\forall m \in M, \quad c(m) = \sum_{u \in \sigma(m)} u \otimes \chi_m(u)$$

It suffices now to apply Theorem I.4 in order to conclude. ■

Note. This result shows that the construction of Schmidt (see [Sm]) is in a certain sense maximal.

II. THE FREE PARTIALLY COMMUTATIVE MONOID

(1) *Quasi-direct Products*

Let M, N be two monoids and let μ, ν be two mappings from $N \times M$ into M and N , respectively. Let us now study how to equip the set $M \times N$ with a monoid structure whose law will be defined for every $m, m' \in M$ and $n, n' \in N$ by

$$(m, n) \cdot (m', n') = (m \cdot \mu(n, m'), \nu(n, m') \cdot n'). \quad (qd\#)$$

An easy computation, which we leave to the reader, shows that the above law is associative iff μ and ν satisfy the following property:

$$\begin{aligned} &\forall m, m', m'' \in M, \forall n, n', n'' \in N, \\ &\left\{ \begin{array}{l} \mu(n, m') \mu(\nu(n, m') n', m'') = \mu(n, m' \mu(n', m'')) \\ \nu(n, m' \mu(n', m'')) \nu(n', m'') = \nu(\nu(n, m') n', m''). \end{array} \right. \quad (\mathcal{A}\mathcal{O}) \end{aligned}$$

Moreover, it is also obvious that $(1_M, 1_N)$ is a unit for the previous law iff μ and ν satisfy the conditions

$$\begin{aligned} \forall n \in N, \forall m \in M, \\ \left\{ \begin{array}{l} \nu(n, 1) = n \\ \mu(n, 1) = 1_M \end{array} \right. \quad \text{and} \quad \left\{ \begin{array}{l} \mu(1, m) = m \\ \nu(1, m) = 1_N. \end{array} \right. \quad (\mathcal{U}_n) \end{aligned}$$

Therefore we can propose the following definition:

DEFINITION II.1. Let M, N be two monoids and let μ, ν be two mappings from $N \times M$ into M and N , respectively, that satisfy the properties $(\mathcal{A}_{\beta\beta})$ and $(\mathcal{U}_n)$. Then the *quasi-direct product* of M and N is the monoid, denoted $M \ltimes_{\mu, \nu} N$, which is defined by the above law (*q.d.p.*).

EXAMPLES. (1) Every semidirect product of monoids is a quasi-direct product. Indeed, let M, N be two monoids and let φ be a monoid morphism from N into $\text{End}(M)$. Then the semidirect product $M \ltimes_{\varphi} N$ is the monoid constructed over $M \times N$ whose law is given by (see [Ei.2] or [Pin] for more details)

$$\forall m \in M, \forall n \in N, \quad (m, n) \cdot (m', n') = (m \cdot \varphi(n)(m'), n \cdot n').$$

Let us define two mappings μ and ν from $N \times M$ into M and N , respectively:

$$\forall (n, m) \in N \times M, \quad \mu(n, m) = \varphi(n)(m) \quad \text{and} \quad \nu(n, m) = n.$$

The reader will easily check that μ and ν satisfy the previous conditions $(\mathcal{A}_{\beta\beta})$ and $(\mathcal{U}_n)$. Hence the semidirect product $M \ltimes_{\varphi} N$ is clearly isomorphic to the quasi-direct product $M \ltimes_{\mu, \nu} N$.

(2) Let (X, Y) be a bisection of a free monoid A^* (see [Vi.2] or [BePe, VII.5] for more details). This means that X and Y are codes over A such that the free monoid A^* has the following factorization: $A^* = X^*Y^*$. It clearly follows that $Y^*X^* \subset X^*Y^*$. Thus let us consider the two mappings μ and ν from $Y^* \times X^*$ into X^* and Y^* , respectively, that are defined by

$$\forall y \in Y^*, \forall x \in X^*, \quad y \cdot x = \mu(y, x) \cdot \nu(y, x).$$

The reader will easily check that conditions $(\mathcal{A}_{\beta\beta})$ and $(\mathcal{U}_n)$ are satisfied. Hence it follows from Definition II.1 that the free monoid A^* is isomorphic to the following quasi-direct product of free monoids $X^* \ltimes_{\mu, \nu} Y^*$.

(2) *Lazard's Partially Commutative Bisection*

Before stating our main result, let us give the following definition:

DEFINITION II.2. Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$ and let $w \in M(A, \mathcal{G})$. Then we call *terminal alphabet*⁴ of w the subset of A , denoted by $\text{TAIph}(w)$, defined as follows:

$$\text{TAIph}(w) = \{a \in A, w \in M(A, \mathcal{G}) \cdot a\}.$$

Criterion. Let $w = a_1 \cdots a_n$ be an element of $M(A, \mathcal{G})$. Then a letter a of A belongs to $\text{TAIph}(w)$ if and only if a is a letter of w for which there exists an index $i \in \llbracket 1, n \rrbracket$ such that $a = a_i$ and $(a_j, a) \in \mathcal{G}$ for every $j \in \llbracket i+1, n \rrbracket$.

Notation. Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$ and let $B \subset A$ such that $Z = A - B$ is *non-commutative* for $\mathcal{G}$. Then we denote by $C_Z(A)$ the following subset of $M(A, \mathcal{G})$:

$$C_Z(A) = \{w \cdot z, w \in \langle B \rangle, z \in Z, \text{TAIph}(wz) = \{z\}\}.$$

Notes. (1) $\langle B \rangle$ denotes the submonoid of $M(A, \mathcal{G})$ generated by B .

(2) When there is no ambiguity, we just denote $C_Z(A)$ by C_Z .

We can now give the following result which gives a new normal form for the elements of $M(A, \mathcal{G})$:

PROPOSITION II.1. Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$ and let B be a subalphabet of A such that $Z = A - B$ is non-commutative for $\mathcal{G}$. Then every word w of $M(A, \mathcal{G})$ has a unique decomposition as follows:

$$w = c_1 \cdots c_n m \quad \text{with} \quad c_i \in C_Z \quad \text{for every } i \text{ and } m \in \langle B \rangle. \quad (\mathcal{D}_{ec})$$

Proof. We prove independently the existence and the uniqueness of the decomposition $(\mathcal{D}_{ec})$ under the above hypotheses.

(a) *Existence of the decomposition.* To prove the existence of $(\mathcal{D}_{ec})$, we use induction on $|w|$. If $|w| \leq 1$, the result is clear since w belongs then either to $Z \subset C_Z$ or to B . Let us suppose it has been proved for $|w| < n$ with $n \geq 2$ and let w be of length n . Then, if α denotes a terminal letter of w , we immediately have by the induction hypothesis

$$w = c_1 \cdots c_n m \alpha, \quad \text{where} \quad m \in \langle B \rangle \text{ and } c_i \in C_Z \text{ for every } i.$$

If $n \geq 1$, the result follows easily from the induction hypothesis applied to

⁴ This subalphabet of A is also denoted $\text{Max}(w)$ in [CMZ].

$m \cdot \alpha$. The conclusion is also clear when $\alpha \in B$. Thus it remains only to study the case $n=0$ and $\alpha \in Z$, that is to say, $w = m \cdot \alpha$ with $m \in \langle B \rangle$ and $\alpha \in Z$. If $\text{TA}(\text{alph}(m \cdot \alpha)) = \{\alpha\}$, $w \in C_Z$ and the result clearly follows. In the other case, $\text{TA}(\text{alph}(m \cdot \alpha)) \cap \text{Alph}(m) \neq \emptyset$ contains a letter b which occurs in m and which is consequently in B . Then we have $w = m' \cdot b$, where $b \in B$ and $m' \in M(A, \mathcal{G})$. It suffices now to apply the induction hypothesis to m' in order to conclude.

(b) *Uniqueness of the decomposition.* We prove the uniqueness of $(\mathcal{D}ec)$ by induction on $|w|$. When $|w| \leq 1$, there is clearly nothing to prove. Let us now suppose the uniqueness of $(\mathcal{D}ec)$ proved for $|w| < n$, where $n \geq 2$. Then let w be in $M(A, \mathcal{G})$ with $|w| = n$ such that:

$$w = c_1 \cdots c_p m = c'_1 \cdots c'_q m' \quad \text{where } c_i, c'_j \in C_Z \text{ and } m, m' \in \langle B \rangle$$

If $q=0$, w belongs to $\langle B \rangle$. Then no letter of Z can occur in w . It follows immediately that $p=0$ and that $m=m'=w$. This ends the proof in this case. Let us suppose now that $q \geq 1$. Then the previous argument shows that $p \geq 1$. Let us now study the two following distinct cases:

(α) $m \neq 1$. Then let $\alpha \in B$ be a terminal letter of m . Since α is also a terminal letter of w , it must occur in some c'_i or in m' . If α did not occur in m' , there would exist a greatest i such that $c'_i = uv \cdot z$ with $u, v \in \langle B \rangle$ and $\text{TA}(\text{alph}\{c'_i\}) = \{z\}$. Then α would commute with every letter of v and with z . Thus α would belong to $\text{TA}(\text{alph}(c'_i))$ which is not possible. It follows that α occurs in m' and is consequently a terminal letter of m' . Therefore we can write

$$w = c_1 \cdots c_p n \alpha = c'_1 \cdots c'_q n' \alpha \quad \text{with some } n, n' \in \langle B \rangle.$$

Since $M(A, \mathcal{G})$ is cancellative, we have $c_1 \cdots c_p n = c'_1 \cdots c'_q n'$. It is now straightforward to conclude with the induction hypothesis.

(β) $m = 1$. It follows clearly from the previous case that we necessarily have $m' = 1$, that is to say, we have

$$w = c_1 \cdots c_p = c'_1 \cdots c'_q,$$

where $c_p = uz$ and $c_q = u'z'$ with $u, u' \in \langle B \rangle$ and $z, z' \in Z$. Since Z is supposed to be non-commutative, it follows easily that $z = z'$. Hence, since $M(A, \mathcal{G})$ is cancellative, we can write

$$c_1 \cdots c_{p-1} u = c'_1 \cdots c'_{q-1} u'.$$

Using the induction hypothesis, it is now easy to conclude. Thus we have shown the uniqueness of $(\mathcal{D}ec)$ in this last case. The proof is now complete. ■

COROLLARY II.2. *Let A be an alphabet equipped with a partial commutation relation $\mathfrak{g}$ and let B be a subalphabet of A such that $Z = A - B$ is non-commutative for $\mathfrak{g}$. Then the submonoid C_Z^* of $M(A, \mathfrak{g})$ generated by C_Z is free of basis C_Z .*

Note. The above result allows us to speak about C_Z as a Z -code.

Let A be an alphabet equipped with a partial commutation relation $\mathfrak{g}$, let B be a subalphabet of A such that $Z = A - B$ is non-commutative for $\mathfrak{g}$, and let $\mathfrak{g}_B$ be the restriction of $\mathfrak{g}$ to B . Then it follows from Corollary II.2 that the free monoid constructed on C_Z embeds into $M(A, \mathfrak{g})$. The free partially commutative monoid $M(B, \mathfrak{g}_B)$ is also clearly isomorphic to $\langle B \rangle$. Hence, according to these natural identifications and to Proposition II.1, we can define in a unique way two mappings μ, ν from $C_Z^* \times M(B, \mathfrak{g}_B)$ into C_Z^* and $M(B, \mathfrak{g}_B)$ respectively, as follows:

$$\forall c \in C_Z^*, \forall m \in M(B, \mathfrak{g}_B), \quad m \cdot c = \mu(c, m) \cdot \nu(c, m).$$

We can now present Proposition II.1 in the following way:

COROLLARY II.3. *With the previous notations, the monoid $M(A, \mathfrak{g})$ is isomorphic to the following quasi-direct product of the free monoid over C_Z and of the free partially commutative monoid $M(B, \mathfrak{g}_B)$:*

$$M(A, \mathfrak{g}) \simeq C_Z^* \ltimes_{\mu, \nu} M(B, \mathfrak{g}_B).$$

Proof. This follows immediately from Proposition II.1 and Definition II.1. ■

Note. The previous decomposition is not in general a semidirect product.

Proposition II.1 can also be expressed as giving a bisection (cf. [Vi.2] or [BePe, VII.5]) of $M(A, \mathfrak{g})$ which will permit us to extend Lazard's elimination process (see [Vi.2]) to the partially commutative case. Indeed we have:

COROLLARY II.4. *Let A be an alphabet equipped with a partial commutation relation $\mathfrak{g}$, let B be a subalphabet of A such that $Z = A - B$ is non-commutative for $\mathfrak{g}$, and let $\mathfrak{g}_B$ be the restriction of $\mathfrak{g}$ to B . Then the free partially commutative monoid $M(A, \mathfrak{g})$ has the following factorization:*

$$\underline{M(A, \mathfrak{g})} = \underline{(C_Z)^*} \underline{M(B, \mathfrak{g}_B)}. \quad (\mathcal{Bis})$$

Proof. The corollary follows clearly from Proposition II.1. ■

Note. Here the factorization ($\mathcal{B}is$) can be interpreted as a relation between characteristic series in $\mathbb{N}\langle\langle A, \mathcal{G} \rangle\rangle$.

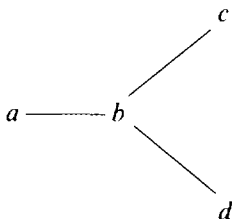
EXAMPLES. (1) When $\mathcal{G} \neq \emptyset$, $M(A, \mathcal{G})$ is the free monoid over A . Hence every subset of A is non-commutative and we clearly have $C_Z = B^* \cdot (A - B)$ for every $B \subset A$. Therefore our construction gives us just the classical Lazard's bisection:

$$A^* = (B^* \cdot (A - B))^* B^*.$$

(2) When $\mathcal{G} = A \times A$, $M(A, \mathcal{G})$ is the free commutative monoid $A^\oplus$ over A . Hence the only subsets of $A^\oplus$ which are non-commutative are the letters and we clearly have $C_Z = \{a\}$ for $B = A - \{a\}$. It follows that our construction gives us just the following natural decomposition of $A^\oplus$:

$$A^\oplus = a^* (A - \{a\})^\oplus.$$

(3) Let $A = \{a, b, c, d\}$ equipped with the partial commutation $\mathcal{G}$ defined by



We can take for instance $B = \{b\}$ and $Z = \{a, c, d\}$, which is non-commutative. Then we clearly have $C_Z = Z$. Therefore $M(A, \mathcal{G})$ can be factorized into two free monoids as follows:

$$M(A, \mathcal{G}) = (a + c + d)^* b^*.$$

(3) Decompositions of $M(A, \mathcal{G})$

We end this section with the following corollaries which show that every free partially commutative monoid can be obtained by using constructions that involve only free monoids. To this end, we need

DEFINITION II.3. Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$. Then a set $\mathcal{F}$ of subsets of A is said to be an *elimination* iff:

- (1) $\mathcal{F}$ is well ordered for the inclusion relation.
- (2) $A \in \mathcal{F}$ and $T(\mathcal{F}) = \bigcap_{X \in \mathcal{F}} X \in \mathcal{F}$.
- (3) For every X in $\mathcal{F} - \{T(\mathcal{F})\}$, $X - X^-$ is non-commutative where $X^- = \bigcup_{Y \not\supseteq X \in \mathcal{F}} Y$.

The elimination is said to be *total* when $T(\mathcal{F}) = \emptyset$.

The following result show that total eliminations always exist.

PROPOSITION II.5. *Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$. Then there exists a total elimination for $(A, \mathcal{G})$.*

Proof. We can order the set $\mathcal{E}$ of eliminations by the relation $\mathcal{F} < \mathcal{G}$ iff $\mathcal{F}$ is a segment of $\mathcal{G}$ (see [Bo.1, III.16]). The reader will easily check that $(\mathcal{E}, <)$ is inductively ordered. Hence it follows from Zorn's lemma that $\mathcal{E}$ has a maximal element $\mathcal{F}$, which is in fact a total elimination as it can be easily proved. ■

Note. When A is a finite alphabet, a total elimination is just a decreasing sequence $(A_k)_{k=1, n+1}$ of subsets of A such that $A_1 = A$, $A_{n+1} = \emptyset$ and $A_k - A_{k+1}$ is non-commutative for every k in $\llbracket 1, n \rrbracket$. Observe that the existence of a total elimination is here obvious since we can eliminate just one letter at each step of the elimination, for instance.

PROPOSITION II.6. *Let A be a finite alphabet equipped with a partial commutation relation $\mathcal{G}$. Then, for every total elimination $\mathcal{F} = (A_k)_{k=1, n+1}$ for $(A, \mathcal{G})$, the free partially commutative monoid $M(A, \mathcal{G})$ is isomorphic to a quasi-direct product of free monoids of the form*

$$M(A, \mathcal{G}) \simeq (C_{A_1 - A_2}(A_1))^* \ltimes ((C_{A_2 - A_3}(A_2))^* \ltimes (\dots \ltimes (C_{A_n - A_{n+1}}(A_n))^* \dots)).$$

Proof. Our result follows clearly from Corollary II.3 by induction on $|A|$. ■

PROPOSITION II.7. *Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$. Then, for every total elimination $\mathcal{F}$ for $(A, \mathcal{G})$, the free partially commutative monoid $M(A, \mathcal{G})$ has the following factorization:*

$$M(A, \mathcal{G}) = \prod_{X \in \mathcal{F}} C_{X - X^-}(X)^*.$$

Proof. Our proposition follows easily from Corollary II.4 by using a transfinite induction. ■

Remarks. (1) When A is finite, the previous proposition also gives new normal forms for the elements of the free partially commutative monoid $M(A, \mathcal{G})$.

(2) Every free monoid A^* has a complete factorization (see [Vi.2] or [BePe, VIII.4, pp. 360–363]). This means that we can factorize A^* as

$$A^* = \prod_{i \in I} w_i^*,$$

where $(w_i)_{i \in I}$ is a family of words of A^* , indexed by a totally ordered set I . A classical complete factorization is the Lyndon factorization, for instance (cf. [Vi.2] or [BePe, VII.4]). Using this result, Proposition 0.2, and Proposition II.7, it is now obvious that a complete factorization, i.e., a factorization of the form $(w_i^*)_{i \in I}$, but where the w_i are here elements of $M(A, \mathcal{G})$, can be constructed on any free partially commutative monoid $M(A, \mathcal{G})$.

EXAMPLES. (1) Let us consider an alphabet $A_n = \{a_1, \dots, a_n\}$ equipped with a partial commutation relation λ_n defined by

$$a_i a_j \equiv_{\lambda_n} a_j a_i \quad \text{for every } |i - j| \geq 2.$$

It is more convenient to represent this commutation relation here by its non-commutation graph $\bar{\lambda}_n$, which is very simple:

$$a_1 \text{ --- } a_2 \text{ --- } a_3 \text{ --- } \dots \text{ --- } a_n.$$

An easy application of Corollary II.4 (see also [DK.4]) permits us to show that we have the following relation:

$$C_{\{a_n\}}(A_n) = C_{\{a_{n-1}\}}(A_{n-1})^* a_n.$$

It follows immediately from this inductive relation that we have

$$C_{\{a_n\}}(A_n) = (((\dots((a_1^* a_2)^* a_3)^* \dots)^* a_n)^*. \quad (\mathcal{C})$$

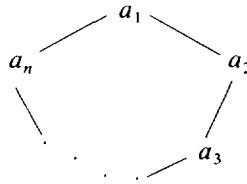
Using Corollary II.4, it is now straightforward to show that the free partially commutative $M(A_n, \lambda_n)$ has the following factorization:

$$\underline{M(A_n, \lambda_n)} = \prod_{k=n}^1 (((\dots((a_1^* a_2)^* a_3)^* \dots)^* a_k)^*).$$

(2) Let us consider again an alphabet $A_n = \{a_1, \dots, a_n\}$ equipped with a partial commutation relation γ_n defined by

$$a_i a_j \equiv_{\gamma_n} a_j a_i \quad \text{for every } i - j \neq \pm 1[n].$$

Again it is more convenient to represent this commutation relation by its non-commutation graph $\bar{\gamma}_n$:



An easy computation based on Corollary II.4 (see also [DK.4]) permits us to show that the following relation holds:

$$C_{\{a_n\}}(A_n) = C_{\{a_1, a_{n-1}\}}(A_{n-1}) * a_n = [C_{\{a_1\}}(A_{n-1}) \cup C_{\{a_{n-1}\}}(A_{n-1})] * a_n.$$

Observe now that the submonoid of $M(A_n, \gamma_n)$ generated by A_{n-1} is isomorphic to the free partially commutative monoid $M(A_{n-1}, \lambda_{n-1})$ of the last example. Thus it follows easily from the computations made in the previous example and from the above relation that we have

$$\begin{aligned} C_{\{a_n\}}(A_n) = & (((\dots((a_1^* a_2)^* a_3)^* \dots a_{n-2})^* a_{n-1} \\ & \cup ((\dots((a_{n-1}^* a_{n-2})^* a_{n-3})^* \dots a_2)^* a_1)^* a_n. \end{aligned}$$

Applying Corollary II.4, it is now immediate to obtain the following factorization of the free partially commutative monoid $M(A_n, \gamma_n)$:

$$\underline{M(A_n, \gamma_n)} = \underline{C_{\{a_n\}}(A_n)}^* \prod_{k=n-1}^1 \underline{((\dots((a_1^* a_2)^* a_3)^* \dots)^* a_k)^*}.$$

Remark. Let $n \geq 1$ and let $\mathfrak{S}_{n+1}$ denote the symmetric group of order $n+1$. The Coxeter presentation of $\mathfrak{S}_{n+1}$ is defined as

$$\mathfrak{S}_{n+1} \simeq \langle (t_i)_{i=1, n}; t_i t_j = t_j t_i \text{ for } |i-j| \geq 2, (t_i t_{i+1})^3 = 1 \text{ for } i \geq 1 \rangle$$

(see [Bo.6, IV] or [CoMo] for more details), where t_i stands in fact for the transposition $(i, i+1)$. Let us consider again the monoid $M(A_n, \lambda_n)$ of the first above example; we can clearly define an epimorphism of monoids φ from $M(A_n, \lambda_n)$ onto $\mathfrak{S}_{n+1}$ such that $\varphi(a_i) = t_i$ for every $i \geq 1$. Observe that in fact we have the relation $\varphi(C_{\{a_n\}}^*) = \mathfrak{S}_{n+1}$. Note also that it can be shown that the subset $\mathcal{Sect}$ of $C_{\{a_n\}}^*$ defined by

$$\mathcal{Sect} = \{a_1^{\epsilon_1} (a_1 a_2)^{\epsilon_2} \dots (a_1 \dots a_n)^{\epsilon_n}, \epsilon_i \in \llbracket 1, i \rrbracket \text{ for every } i\}$$

is a section of the morphism φ .

(4) Optimality of Lazard's Factorization

We study in the section the optimality of the factorization of $M(A, \vartheta)$ which is given by Prop. II.7 when A is finite. To this end, let us at first recall the following definition (see [Bg] for more details):

DEFINITION II.4. Let $G = (X, V)$ be a graph. Then the *chromatic number* $\gamma(G)$ is the minimum number of colours which are needed to colour the vertices of G in such a way that two adjacent vertices are not in the same colour.

Notation. When ϑ is a partial commutation relation on a finite alphabet A , we denote by $\gamma(\vartheta)$ the chromatic number of the commutation graph of ϑ .

Observe first that there exists a total elimination for (A, ϑ) with $\gamma(\vartheta)$ non-empty elements. Indeed, according to the definition of $\gamma(\vartheta)$, there exists a partition $(\mathcal{A}_i)_{i=1, \gamma(\vartheta)}$ of the alphabet A such that

$$\forall i \in \llbracket 1, \gamma(\vartheta) \rrbracket, a \in \mathcal{A}_i \text{ and } b \in \mathcal{A}_j \Rightarrow (a, b) \notin \vartheta.$$

Let us now define for every $i \in \llbracket 1, \gamma(\vartheta) + 1 \rrbracket$ the set

$$A_i = \bigcup_{k=i}^{\gamma(\vartheta)} \mathcal{A}_k.$$

Then it is clear that the family $(A_i)_{i=1, \gamma(\vartheta)+1}$ is a total elimination for (A, ϑ) with $\gamma(\vartheta)$ non-empty elements. It follows immediately that there exists an associated factorization of $M(A, \vartheta)$ involving exactly $\gamma(\vartheta)$ free monoids which is given by Proposition II.7. Moreover such a factorization is optimal in a sense that is made precise by the following theorem:

THEOREM II.8. Let A be a finite alphabet equipped with a partial commutation relation ϑ and let us consider a factorization of $M(A, \vartheta)$ in free monoids,

$$\underline{M(A, \vartheta)} = \prod_{i=1}^N \underline{X_i^*}. \quad (*)$$

Then the number N of free monoids involved in $(*)$ is greater than or equal to the chromatic number $\gamma(\vartheta)$ of the commutation graph of ϑ .

Proof. Let us consider a finite factorization of $M(A, \vartheta)$ into N free monoids as above. Then let us consider the set $I = \{i \in \llbracket 1, N \rrbracket,$

$X_i \cap A \neq \emptyset$. Observe first that it follows by an immediate length argument that $(A \cap X_i)_{i \in I}$ is a partition of A . Let now $i \in I$. Then we have

$$a \neq b \in X_i \cap A \Rightarrow (a, b) \notin \mathcal{G}. \quad (1)$$

Indeed, if this were the case, we would have $ab = ba$. But, this relation occurs in the free monoid X_i^* : it follows that there exists $x \in X_i^*$ such that $a = x^n$ and $b = x^m$ for some n, m in $\mathbb{N}$ (see [Lo]). Taking the length in $M(A, \mathcal{G})$ of these two relations, it follows clearly that $n = m = 1$ and therefore that $a = x = b$. This contradiction ensures that property (1) holds.

Then, if we colour the letters of $X_i \cap A$ in colour i , two adjacent vertices of the communication graph of $\mathcal{G}$ are not in the same colour. It follows that

$$\gamma(\mathcal{G}) \leq |I| \leq N.$$

This last relation proves the second assertion of our theorem. Therefore it ends our proof. ■

Remark. The above results give us the optimal way to factorize $M(A, \mathcal{G})$ into free monoids: indeed it suffices to colour the commutation graph of $\mathcal{G}$ with $\gamma(\mathcal{G})$ colours in such a way that any two adjacent letters are not in the same colour. Then we can apply our method by eliminating successively the letters of the same colour in order to obtain the optimal factorization. However it is to be noted that other factorizations can sometimes be preferred to the optimal one (for symmetry reasons for instance).

III. THE FREE PARTIALLY COMMUTATIVE LIE ALGEBRA

(1) Definition

Let us recall the definition of the free partially commutative Lie algebra that was already studied in [DK.1, Du]:

DEFINITION III.1. Let K be a ring and let A be an alphabet equipped with a commutation relation $\mathcal{G}$. Then the *free partially commutative Lie algebra* is the Lie algebra, denoted $L_K(A, \mathcal{G})$ or $L(A, \mathcal{G})$ if there is no ambiguity, that is defined by the following presentation:

$$L_K(A, \mathcal{G}) = \langle A; [a, b], (a, b) \in \mathcal{G} \rangle.$$

Notes. (1) This means that if $I_{\mathcal{G}}$ denotes the Lie ideal of $L(A)$ generated by the family $\{[a, b], (a, b) \in \mathcal{G}\}$, then $L_K(A, \mathcal{G})$ is exactly the following quotient Lie algebra: $L_K(A, \mathcal{G}) = L(A)/I_{\mathcal{G}}$.

(2) $L(A, \mathcal{G})$ is characterized by the following universal property: for every mapping f from A into a Lie algebra $\mathcal{G}$ such that

$$(a, b) \in \mathcal{G} \Rightarrow [f(a), f(b)] = 0,$$

there exists a unique Lie algebra morphism $\tilde{f}$ from $L(A, \mathcal{G})$ into $\mathcal{G}$ which extends f , i.e., such that the following diagram is commutative,

$$\begin{array}{ccc} A & \xrightarrow{i} & L(A, \mathcal{G}) \\ f \downarrow & \searrow \tilde{f} & \\ \mathcal{G} & & \end{array}$$

where i denotes the natural injection from A into $L(A, \mathcal{G})$.

(3) Let us recall that the enveloping algebra of $L_K(A, \mathcal{G})$ is isomorphic to the K -algebra $K\langle A, \mathcal{G} \rangle = K\langle A; \equiv_{\mathcal{G}} \rangle$ (see [Du] or [DK.1]).

(4) For every letter $a \in A$, we can define the derivation $\text{ad } a$ of $L(A, \mathcal{G})$ that maps every $x \in L(A, \mathcal{G})$ on $\text{ad } a \cdot x = [a, x]$. Hence we can consider the mapping from A into $\text{End}_K(L(A, \mathcal{G}))$ which associates with every $a \in A$ the endomorphism $\text{ad } a$. This mapping is compatible with the commutations thanks to the relation (derived of Jacobi's identity) $[\text{ad } a, \text{ad } b] = \text{ad}[a, b] = 0$ when $(a, b) \in \mathcal{G}$. According to the universal property of $M(A, \mathcal{G})$, it follows that we can define for every element w of $M(A, \mathcal{G})$ an endomorphism $\text{ad } w$ of $L(A, \mathcal{G})$ such that

$$\text{ad } w = \text{ad } a_1 \circ \cdots \circ \text{ad } a_n \quad \text{when} \quad w = a_1 \cdots a_n.$$

Moreover we can also consider the natural K -linear extension of ad to $K\langle A, \mathcal{G} \rangle$ since we have $K\langle A, \mathcal{G} \rangle \simeq K[M(A, \mathcal{G})]$. It is then easy to check that

$$P \in L(A, \mathcal{G}) \Rightarrow \text{ad } P \in \text{Der}(L(A, \mathcal{G})).$$

Moreover we have $\text{ad } P \cdot Q = [P, Q] = PQ - QP$ for every Q in $K\langle A, \mathcal{G} \rangle$.

(2) The Partially Commutative Lazard's Elimination Theorem

Let A be an alphabet equipped with a commutation relation $\mathcal{G}$ and let B be a subalphabet of A such that $Z = A - B$ is non-commutative. Let us consider the free Lie algebra $L(C_Z)$ build over the associated Z -code C_Z . For every b in B , let us now define the mapping D_b from C_Z into $L(C_Z)$ by

$$D_b(c) = \begin{cases} bc & \text{if } bc \in C_Z \\ 0 & \text{otherwise} \end{cases}$$

for every c in C_Z . Let us also denote by D_b the unique extension of D_b as a derivation of $L(C_Z)$ (see [Bo.5, II, Sect. 2, No. 8]). Hence we construct a mapping D from B into the Lie algebra $\text{Der}(L(C_Z))$ of the derivations of $L(C_Z)$:

$$D: B \rightarrow \text{Der}(L(C_Z))$$

$$b \rightarrow D_b.$$

Let us now show that this mapping can be extended onto a Lie algebra morphism of $L(B, \mathfrak{g}_B)$ into $\text{Der}(L(C_Z))$. This is the purpose of the proposition:

PROPOSITION III.1. *There is a unique Lie morphism of $L(B, \mathfrak{g}_B)$ into $\text{Der}(L(C_Z))$ that extends D .*

Proof. According to the universal property of $L(B, \mathfrak{g}_B)$, it suffices to show that $[D_b, D_{b'}] = 0$ when $(b, b') \in \mathfrak{g}_B$. Since C_Z is a generating system of $L(C_Z)$, it suffices in fact to show that

$$[D_b, D_{b'}](c) = D_b D_{b'}(c) - D_{b'} D_b(c) = 0. \quad (1)$$

for every c in C_Z when $(b, b') \in \mathfrak{g}$. Two situations are now to be considered:

(a) bc and $b'c$ are in C_Z . Then we have

$$[D_b, D_{b'}](c) = D_b(b'c) - D_{b'}(bc). \quad (2)$$

Since b and b' commute, we have $bb'c = b'bc$. Hence, $bb'c \in C_Z$ iff $b'bc \in C_Z$. It follows easily now that $D_b(b'c) - D_{b'}(bc)$ is always equal to 0.

(b) bc or $b'c$ is not in C_Z . If bc and $b'c$ are not in C_Z , we clearly have $[D_b, D_{b'}](c) = 0$. By symmetry, we can restrict ourselves now to consider only the case $bc \notin C_Z$ and $b'c \in C_Z$. Then we have

$$[D_b, D_{b'}](c) = D_b(b'c). \quad (3)$$

According to the definition of C_Z , we have $c = w \cdot z$ with $w \in \langle B \rangle$, $z \in Z$, and $\text{TA}(\text{Alph}(wz)) = \{z\}$. Since $bc = bw \cdot z \notin C_Z$, there exists a terminal letter β of bc that is different from z . It follows that $bc = w'\beta$ for some $w' \in M(A, \mathfrak{g})$. Since b and b' commute, we can write $bb'c = b'bc = b'w'\beta$. Hence $\beta \neq z \in \text{TA}(\text{Alph}(bb'c))$. Since β is necessarily in B , it follows that $bb'c \notin C_Z$. Hence it follows from (3) that $[D_b, D_{b'}](c) = 0$. Thus we showed that (1) holds in every case. Our proof is now finished. ■

The Lie morphism from $L(B, \mathfrak{g}_B)$ into $\text{Der}(L(C_Z))$ introduced in the previous proposition will also be denoted D on now on. Using the

universal property of $L(C_Z)$, we can clearly define a Lie algebra morphism α from $L(C_Z)$ into $L(A, \mathfrak{g})$ such that we have for every c in C_Z

$$\alpha(c) = \text{ad } w \cdot z, \quad \text{where } c = w \cdot z, w \in \langle B \rangle, z \in Z, \text{ and } \text{TAlph}(c) = \{z\}.$$

The following proposition gives us a useful property of α :

PROPOSITION III.2. *For every $p \in L(C_Z)$ and $q \in L(B, \mathfrak{g}_B)$, we have*

$$\alpha(D_q(p)) = [q, \alpha(p)]. \quad (*)$$

Proof. Let $\mathcal{Q}$ denote the subset of $L(B, \mathfrak{g}_B)$ consisting of all $q \in L(B, \mathfrak{g}_B)$ that satisfy $(*)$ for every $p \in L(C_Z)$. Using Jacobi's identity, it is easy to see that $\mathcal{Q}$ is in fact a Lie subalgebra of $L(B, \mathfrak{g}_B)$. Therefore it suffices to prove that $B \subset \mathcal{Q}$ in order to obtain $\mathcal{Q} = L(B, \mathfrak{g}_B)$. Hence let b be in B . We must prove that we have, for every p in $L(C_Z)$,

$$\alpha(D_b(p)) = [b, \alpha(p)]. \quad (**)$$

Let us now consider $\mathcal{P}$ the subset of $L(C_Z)$ that consists of all $p \in L(C_Z)$ that satisfies $(**)$. Using again Jacobi's identity, it is easy to prove that $\mathcal{P}$ is in fact a Lie subalgebra of $L(C_Z)$. It follows that it suffices to show $(**)$ on C_Z in order to conclude that $\mathcal{P} = L(C_Z)$. Thus let $c = wz$ be in C_Z , where $w \in \langle B \rangle$, $z \in Z$, and $\text{TAlph}(c) = \{z\}$. We must prove that

$$\alpha(D_b(w \cdot z)) = [b, \alpha(w \cdot z)] = [b, \text{ad } w \cdot z] = \text{ad } bw \cdot z. \quad (0)$$

If $bwz \in C_Z$, this result follows immediately from our definitions. But, if $bwz \notin C_Z$, we must show that $\text{ad } bw \cdot z = 0$ in order to obtain (0). Observe that b necessarily commutes here with w and z . It follows that we have

$$\begin{aligned} \text{ad } bw \cdot z &= \text{ad } b \circ \text{ad } w \cdot z = \text{ad } w \circ \text{ad } b \cdot z + [\text{ad } b, \text{ad } w] \cdot z \\ &= \text{ad } w \cdot [b, z] + \text{ad}[b, w] \cdot z = 0. \end{aligned}$$

Therefore our proposition follows. ■

Let us now consider the mapping φ from $L(C_Z) \ltimes_D L(B, \mathfrak{g}_B)$ into $L(A, \mathfrak{g})$ that is defined for every $p \in L(C_Z)$ and $q \in L(B, \mathfrak{g}_B)$ by

$$\varphi(p, q) = \alpha(p) + q.$$

We can now give our main result, which generalizes Lazard's elimination theorem for free Lie algebras (see [Bo.5, II, Sect. 2, No. 9] or [La]). It proves that the mapping φ is a Lie algebra isomorphism between $L(A, \mathfrak{g})$ and $L(C_Z) \ltimes_D L(B, \mathfrak{g}_B)$.

THEOREM III.3 (Lazard's Partially Commutative Elimination). *Let A be an alphabet equipped with a commutation relation ϑ and let B be a sub-alphabet of A such that $Z = A - B$ is non-commutative. Then the mapping φ is an isomorphism from the free partially commutative Lie algebra $L(A, \vartheta)$ into the following semidirect product of Lie algebras:*

$$L(A, \vartheta) \simeq L(C_Z) \ltimes_D L(B, \vartheta_B).$$

Proof. Let us first prove that φ is a morphism of Lie algebras. Indeed we have, for every p, p' in $L(C_Z)$ and q, q' in $L(B, \vartheta_B)$,

$$\begin{aligned} [\varphi(p, q), \varphi(p', q')] &= [\alpha(p), \alpha(p')] - [q', \alpha(p)] + [q, \alpha(p')] + [q, q'] \\ &= \alpha([p, p'] - D_{q'}(p) + D_q(p')) + [q, q'] \\ &= \varphi([(p, q), (p', q')]), \end{aligned}$$

according to Proposition III.2 and Definition 0.4. To prove that φ is an isomorphism, we construct the inverse of φ . Let us now consider the mapping ψ from A into $L(C_Z) \ltimes_D L(B, \vartheta_B)$ that is defined by

$$\forall a \in A, \psi(a) = \begin{cases} (a, 0) & \text{if } a \in Z \\ (0, a) & \text{if } a \in B. \end{cases}$$

Let us prove that we have

$$(a, b) \in \vartheta \Rightarrow [\psi(a), \psi(b)] = 0. \quad (0)$$

Let $a, b \in A$ such that $(a, b) \in \vartheta$. To prove (0), we must study four cases. First, if $a, b \in B$, we have $[\psi(a), \psi(b)] = [(0, a), (0, b)] = (0, [a, b]) = (0, 0)$. Second, if $a \in Z$ and $b \in B$, we have $[\psi(a), \psi(b)] = [(a, 0), (0, b)] = (-D_b(a), 0) = (0, 0)$. The case $a \in B$ and $b \in Z$ is similar to the previous one and the case $a, b \in Z$ cannot occur here because Z is non-commutative. Hence (0) holds. Therefore, using the universal property of $L(A, \vartheta)$, we can extend ψ to a Lie morphism from $L(A, \vartheta)$ into $L(C_Z) \ltimes_D L(B, \vartheta_B)$ that we still denote by ψ .

Let us now introduce the subset $A' = C_Z \times \{0\} \cup \{0\} \times B$ of $L(C_Z) \ltimes_D L(B, \vartheta_B)$. The reader will easily check that A' is a generating family of $L(C_Z) \ltimes_D L(B, \vartheta_B)$. But it is also immediate to check that $\psi \circ \varphi$ and $\varphi \circ \psi$ are respectively identical on A and A' . It follows that $\psi \circ \varphi$ and $\varphi \circ \psi$ are respectively identical on $L(A, \vartheta)$ and on $L(C_Z) \ltimes_D L(B, \vartheta_B)$. Hence φ is a Lie isomorphism with ψ as inverse. Our theorem is now proved. ■

The elimination theorem can also be "internally" restated as follows:

THEOREM III.4. *Let A be an alphabet equipped with a commutation relation $\mathfrak{g}$, let B be a subalphabet of A such that $Z = A - B$ is non-commutative, and let I_Z be the Lie ideal of $L(A, \mathfrak{g})$ generated by Z . Then we have:*

(i) *I_Z is a free Lie algebra and a basic family of I_Z is*

$$T_Z = \{\text{ad } w \cdot z, w \in \langle B \rangle, z \in Z, \text{TA}(\text{ph}(wz)) = \{z\}\}.$$

(ii) *The K -module $L(A, \mathfrak{g})$ can be decomposed in direct sum as*

$$L(A, \mathfrak{g}) = I_Z \oplus L(B, \mathfrak{g}).$$

Remark. Theorems III.3 and III.4 are just the Lie algebra versions of Corollaries II.3 and II.4. Observe also that, according to our results, every effective description of a Z -code in the free monoid A^* gives immediately an effective description of the basic family T_Z of the free Lie algebra I_Z . This motivates the combinatorial study of the Z -codes: we refer the reader especially to [DK.4] for effective methods of computing Z -codes.

EXAMPLE. Let us consider the alphabet $A = \{a, b, c\}$ equipped with a partial commutation relation $\mathfrak{g}$ defined by the following commutation graph:

$$a \text{ --- } b \qquad c$$

Then $Z = \{c\}$ is a non-commutative subset of A and we clearly have

$$C_Z(A) = \langle a, b \rangle \cdot c = (a + b)^{\oplus} \cdot c = a^* b^* \cdot c.$$

We can apply III.4 in order to obtain

$$L(A, \mathfrak{g}) = L(\{\text{ad } w \cdot c, w \in (a + b)^{\oplus}\}) \oplus L(\{a, b\}, \mathfrak{g}).$$

It is now immediate to obtain the following decomposition of $L(A, \mathfrak{g})$:

$$L(A, \mathfrak{g}) = L(\{a^n b^m \cdot c, n, m \geq 0\}) \oplus K \cdot a \oplus K \cdot b. \quad (\text{Sum})$$

(3) *Decompositions of $L(A, \mathfrak{g})$*

As in Section II.3, we can apply the previous results to obtain interesting decompositions of a free partially commutative Lie algebra into free Lie algebras:

PROPOSITION III.5. *Let A be a finite alphabet equipped with a partial commutation relation $\mathfrak{g}$. For every total elimination $\mathcal{F} = (A_k)_{k=1, n+1}$ for $(A, \mathfrak{g})$, the free partially commutative Lie algebra $L(A, \mathfrak{g})$ is then isomorphic to a semidirect product of free Lie algebras of the form*

$$L(A, \mathfrak{g}) \simeq L(C_{A_1 - A_2}(A_1)) \ltimes (L(C_{A_2 - A_3}(A_2)) \ltimes (\dots \ltimes L(C_{A_n - A_{n+1}}(A_n)) \dots)).$$

Proof. Our result follows immediately from Theorem III.3 by induction on $|A|$. ■

PROPOSITION III.6. *Let A be an alphabet equipped with a partial commutation relation $\mathfrak{g}$, let $\mathcal{F}$ be a total elimination for $(A, \mathfrak{g})$, and let us denote by $\mathfrak{g}_X$ the restriction of $\mathfrak{g}$ to $X \in \mathcal{F}$. Then, for every $X \in \mathcal{F}$, the Lie ideal I_X of $L(X, \mathfrak{g}_X)$ generated by $X - X^-$ is a free Lie algebra of basic family*

$$T_X = \{\text{ad } w \cdot z, w \in \langle A_{X^-} \rangle, z \in X - X^-, \text{TA}(\text{Alph}(wz)) = \{z\}\}.$$

Moreover, the free partially commutative Lie algebra $L(A, \mathfrak{g})$ has the following direct sum decomposition:

$$L(A, \mathfrak{g}) = \bigoplus_{X \in \mathcal{F}} L(T_X).$$

Proof. Our proposition follows immediately from Theorem III.4 by a transfinite induction. ■

Remark. Since a free Lie algebra is a free K -module (see [Bo.5]), it follows from the previous result that every free partially commutative Lie algebra is also a free K -module (see [Du] or [DK.1] for other proofs). Observe also that the previous result gives us a natural way to compute K -bases of the free K -module $L(A, \mathfrak{g})$, since there are several classical methods to obtain a K -basis for a free Lie algebra (see [Bo.5, Vi.2]).

EXAMPLE. Let us take again the example considered at the end of the last subsection. Let us now introduce the lexicographic order on $\mathbb{N} \times \mathbb{N}$ defined by $(1, 0) < (0, 1)$. Then let us order the family $\mathcal{F} = \{\text{ad } a^n b^m \cdot c, n, m \geq 0\}$ by

$$a^n b^m \cdot c < a^{n'} b^{m'} \cdot c$$

$$\text{iff } n + m < n' + m' \text{ or } (n + m = n' + m' \text{ and } (n, m) < (n', m')).$$

We can now compute the classical Lyndon basis (cf. [Vi.2]) of $L(\mathcal{F})$ associated with this ordering of $\mathcal{F}$:

$$\begin{aligned} \mathcal{L}_Y(\mathcal{F}) = \{ & c, [a, c], [b, c], [c, [a, c]], [c, [b, c]], [a, [b, c]], \\ & [a, [a, c]], [b, [b, c]], \dots \}. \end{aligned}$$

According to (*Sum*), it suffices now to add the family $\{a, b\}$ to $\mathcal{L}_Y(\mathcal{F})$ in order to obtain a basis for $L(\{a, b, c\}, \mathfrak{g})$. The reader should compare this basis with the other one obtained in [DK.1] by another decomposition of $L(\{a, b, c\}, \mathfrak{g})$.

(4) *Free Partially Commutative Algebras*

Using the fact that $K\langle A, \vartheta \rangle = K\langle A; \equiv_{\vartheta} \rangle$ is the enveloping algebra of $L(A, \vartheta)$, we can now easily obtain decompositions results for $K\langle A, \vartheta \rangle$:

THEOREM III.7. *Let A be an alphabet equipped with a partial commutation relation ϑ and let $B \subset A$ such that $Z = A - B$ is non-commutative. Then there is a unique isomorphism φ of K -modules from $K\langle C_Z \rangle \otimes K\langle B, \vartheta_B \rangle$ into $K\langle A, \vartheta \rangle$ such that $\varphi(c \otimes b) = cb$ for every $c \in K\langle C_Z \rangle$ and $b \in K\langle B, \vartheta_B \rangle$:*

$$K\langle A, \vartheta \rangle \simeq K\langle C_Z \rangle \oplus K\langle B, \vartheta_B \rangle.$$

Proof. Our result follows from Theorem III.3, from the fact that $K\langle A, \vartheta \rangle$ is the enveloping algebra of $L(A, \vartheta)$, and from Proposition 2.2.10 of [Dx]. ■

Remark. (1) Observe that the above mapping φ is not in general a K -algebra morphism if we take the usual algebra structure on $K\langle C_Z \rangle \otimes K\langle B, \vartheta_B \rangle$. Indeed we do not have in general

$$\varphi((c \otimes b) \cdot (c' \otimes b')) = \varphi(c \otimes b) \cdot \varphi(c' \otimes b') \Leftrightarrow cc'bb' = cbc'b'$$

for every $c, c' \in C_Z^*$ and $b, b' \in M(B, \vartheta_B)$. However, it is easy to prove that there exists a unique K -algebra structure on $K\langle C_Z \rangle \otimes K\langle B, \vartheta_B \rangle$ such that we have for every c, c' in C_Z^* and b, b' in $M(B, \vartheta_B)$

$$(c \otimes b) \cdot (c' \otimes b') = c \cdot \mu(b, c') \otimes \nu(b, c') \cdot b',$$

where $(\mu(b, c'), \nu(b, c'))$ is the factorization of bc' on $C_Z^* \times M(B, \vartheta_B)$ defined by Corollary II.4. It is now easy to show that φ becomes now a K -algebra morphism if we equip $K\langle C_Z \rangle \otimes K\langle B, \vartheta_B \rangle$ with this algebra structure.

(2) When A is a finite alphabet, it follows from the proof of Theorem III.7 and from Proposition III.6 that there exists for every total elimination $(A_i)_{i=1, n+1}$ of (A, ϑ) a unique isomorphism of K -modules φ between the modules

$$K\langle A, \vartheta \rangle \simeq K\langle C_{A_1 \dots A_2}(A_1) \rangle \otimes K\langle C_{A_2 \dots A_3}(A_2) \rangle \otimes \dots \otimes K\langle C_{A_n \dots A_{n+1}}(A_n) \rangle$$

such that $\varphi(c_1 \otimes c_2 \otimes \dots \otimes c_n) = c_1 c_2 \dots c_n$ for every $c_i \in C_{A_i \dots A_{i+1}}$.

(5) *Optimality of Lazard's Decomposition*

As we showed in Section II.4, there exists a total elimination for (A, ϑ) with $\gamma(\vartheta)$ non-empty elements. Hence there exists an associated decomposition of $L(A, \vartheta)$ in a direct sum of $\gamma(\vartheta)$ free Lie algebras which is given by Proposition III.6. In the same way, we have an associated decomposition of

$K\langle A, \mathfrak{g} \rangle$ in a tensor product of $\gamma(\mathfrak{g})$ free associative algebras, given by the second remark following Theorem III.7. Observe also that in all the previous decompositions, every letter of A lies in some of the free objects involved in it.

We can now show that these decompositions are optimal in a certain sense. This is the purpose of the two following results:

THEOREM III.8. *Let K be a domain, let A be a finite alphabet equipped with a partial commutation relation $\mathfrak{g}$ and let us consider a decomposition of the K -module $K\langle A, \mathfrak{g} \rangle$ into a tensor product of free associative K -algebras,*

$$K\langle A, \mathfrak{g} \rangle \simeq K\langle T_1 \rangle \otimes K\langle T_2 \rangle \otimes \cdots \otimes K\langle T_N \rangle, \quad (*)$$

such that the natural K -algebra morphism

$$1 \otimes \cdots \otimes 1 \otimes K\langle T_i \rangle \otimes 1 \otimes \cdots \otimes 1 \rightarrow K\langle A, \mathfrak{g} \rangle$$

is injective and such that every letter of A lies in some $K\langle T_i \rangle$. Then the number N of free associative K -algebras involved in $()$ is greater than or equal to the chromatic number $\gamma(\mathfrak{g})$ of the commutation graph of $\mathfrak{g}$.*

Proof. Let us consider a decomposition of the K -module $K\langle A, \mathfrak{g} \rangle$ into a tensor product of free associative algebras as above. Then let us introduce the set $I = \{i \in [1, N], A \cap K\langle T_i \rangle \neq \emptyset\}$. According to our hypotheses and to the fact that $K\langle T_i \rangle \cap K\langle T_j \rangle = K \cdot 1 \otimes \cdots \otimes 1$ for every $i \neq j$, it follows easily that $(A \cap K\langle T_i \rangle)_{i \in I}$ is a partition of A . Hence, using the same argument as in the proof of Theorem II.8, it suffices to show that we have for every $i \in I$

$$a, b \in A \cap K\langle T_i \rangle \Rightarrow (a, b) \notin \mathfrak{g} \quad (1)$$

in order to conclude our proof. Thus let $a \neq b$ be in $A \cap K\langle T_i \rangle$ for some i in I and let us suppose that $ab = ba$ (2). Then it follows from Bergman's centralizer theorem (cf. [Co, p. 246]) that there exists $f \in K\langle T_i \rangle$ such that

$$a = \sum_{i=0}^n k_i f^i \text{ and } b = \sum_{i=0}^m l_i f^i \quad \text{with } k_i \text{ and } l_i \text{ in } K, \quad (3)$$

where $k_n \neq 0$ and $l_m \neq 0$. But the free monoid T_i^* can be ordered by an order $<$ compatible with the monoid structure (see Corollary III.2 of [DK.2]). It follows that we can write for some $w \in T_i^*$:

$$f = \alpha_w w + \sum_{u < w} \alpha_u u \quad \text{where } \alpha_w \neq 0. \quad (4)$$

Since K is a domain, it follows easily from (3) and (4) that we have $a = w^n$

and $b = w^m$ in the free monoid T_i^* . It follows immediately that $a^m = b^n$ (5). But it follows easily from Corollary 5 of [Bo.3, II, Sect. 3, No. 7] that $K\langle T_i \rangle$ embeds in $K\langle A, \mathfrak{g} \rangle$. Hence relation (5) can be read in $K\langle A, \mathfrak{g} \rangle$. It follows now clearly that $m = n = 1$ and therefore that $a = b$. This contradiction shows that (1) holds. Hence, our proof is finished. ■

COROLLARY III.9. *Let K be a domain, let A be a finite alphabet equipped with a partial commutation relation $\mathfrak{g}$, and let us consider a decomposition of the Lie algebra $L(A, \mathfrak{g})$ into a direct sum of free Lie algebras,*

$$L(A, \mathfrak{g}) \simeq L(T_1) \oplus L(T_2) \oplus \cdots \oplus L(T_N), \quad (*)$$

such that every letter of A lies in some $L(T_i)$. Then the number N of free Lie algebras involved in () is greater or equal to the chromatic number $\gamma(\mathfrak{g})$ of the commutation graph of $\mathfrak{g}$.*

Proof. The corollary follows immediately from Proposition 2.2.10 of [Dx] and from the previous theorem. ■

Remarks. (1) It follows immediately from the previous theorem that the same result holds for decompositions of $L(A, \mathfrak{g})$ in *semidirect products* of free Lie algebras, satisfying the same condition as in Corollary III.9. Thus our method is also optimal in the previous sense for such decompositions.

(2) We conjecture that the condition that every letter of A appears in some component of the considered decompositions is not necessary to state the two previous results; i.e., that a decomposition in $\gamma(\mathfrak{g})$ elements is optimal among all possible decompositions of $K\langle A, \mathfrak{g} \rangle$ or $L(A, \mathfrak{g})$.

IV. THE FREE PARTIALLY COMMUTATIVE GROUP

(1) Definition

Let us recall the definition of the free partially commutative group:

DEFINITION IV.1. Let A an alphabet equipped with a commutation relation $\mathfrak{g}$. Then we call *free partially commutative group* and denote by $F(A, \mathfrak{g})$ the group which is defined by the following presentation:

$$F(A, \mathfrak{g}) = \langle A; aba^{-1}b^{-1}, (a, b) \in \mathfrak{g} \rangle.$$

Notes. (1) Let $F_{\mathfrak{g}}$ denote the normal subgroup of the free group $F(A)$ which is generated by the family of commutators $\{aba^{-1}b^{-1}, (a, b) \in \mathfrak{g}\}$.

Then the above definition just means that the group $F(A, \mathcal{G})$ is defined by $F(A, \mathcal{G}) = F(A)/F_{\mathcal{G}}$.

(2) The free partially commutative group satisfies clearly to the following universal property: for every mapping f from A into a group G such that

$$(a, b) \in \mathcal{G} \Rightarrow f(a)f(b) = f(b)f(a),$$

there exists a unique group morphism $\bar{f}$ from $F(A, \mathcal{G})$ into G that extends f , i.e., such that the following diagram is commutative,

$$\begin{array}{ccc} A & \xrightarrow{i} & F(A, \mathcal{G}) \\ f \downarrow & \searrow \bar{f} & \\ G & & \end{array}$$

where i denotes the natural injection from A into $F(A, \mathcal{G})$.

Observe that the universal property of $M(A, \mathcal{G})$ allows clearly to construct a morphism of monoids ι from $M(A, \mathcal{G})$ into $F(A, \mathcal{G})$ which maps every letter $a \in A$ on itself. The following result says that ι is injective (see also [Ch]):

PROPOSITION IV.1 [Db, p. 66]. *Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$. Then the free partially commutative monoid $M(A, \mathcal{G})$ embeds into the free partially commutative group $F(A, \mathcal{G})$.*

(2) Elimination in $F(A, \mathcal{G})$

Let A be an alphabet equipped with a partial commutation relation $\mathcal{G}$ and let $\bar{A} = \{\bar{a}, a \in A\}$ be a copy of A . For every letter $a \in A \cup \bar{A}$, we denote by $\bar{a}$ the letter $\bar{a}$ when $a \in A$ and the letter b when $a = \bar{b} \in \bar{A}$. Let Θ then be the partial commutation of $A \cup \bar{A}$ which extends $\mathcal{G}$ and is defined by

$$\Theta = \{(a, \bar{a}), a \in A \cup \bar{A}\} \cup \mathcal{G} \cup \{(\bar{a}, b), (a, \bar{b}), (\bar{a}, \bar{b}); (a, b) \in \mathcal{G}\}.$$

Thus we can consider the partially commutative monoid $M(A \cup \bar{A}, \Theta)$. We say that a word w of $M(A \cup \bar{A}, \Theta)$ is *reduced* iff $w \notin M(A \cup \bar{A}, \Theta) \cdot a \cdot \bar{a} \cdot M(A \cup \bar{A}, \Theta)$ for every letter $a \in A \cup \bar{A}$. For every subset $X \subset M(A \cup \bar{A}, \Theta)$, we denote by $\text{Red}(X)$ the subset of X that consists of the reduced words that belong to X . We also denote by π the natural morphism from $M(A \cup \bar{A}, \Theta)$ into $F(A, \mathcal{G})$ defined by

$$\forall a \in A, \quad \pi(a) = a \quad \text{and} \quad \pi(\bar{a}) = a^{-1}.$$

Let now B be a subalphabet of A such that $Z = A - B$ is non-commutative and let $\bar{B} = \{\bar{b}, b \in B\}$. Then we can consider the subset of $M(A \cup \bar{A}, \Theta)$

$$\mathcal{C}_Z(A) = \{w \cdot z, w \in \text{Red}(M(B \cup \bar{B}, \Theta)), z \in Z, \text{TAlph}(w \cdot z) = \{z\}\}.$$

Note that $\mathcal{C}_Z = \mathcal{C}_Z(A)$ generates the submonoid $\langle \mathcal{C}_Z \rangle$ of $M(A \cup \bar{A}, \Theta)$ since it is a subset of the Z -code $C_Z(B \cup \bar{B} \cup Z)$ (Z is clearly a non-commutative subset of $A \cup \bar{A}$). Moreover the structure of $\mathcal{C}_Z$ is closely related to the structure of this Z -code since we have

$$\mathcal{C}_Z(A) = C_Z(B \cup \bar{B} \cup Z) \\ \cap \left(\bigcup_{z \in Z} \left((B \cup \bar{B})^* - \bigcup_{b \in B \cup \bar{B}} (B \cup \bar{B})^* b \bar{b} (B \cup \bar{B})^* \right) z \right).$$

We can now define for every b of $B \cup \bar{B}$ an endomorphism, denoted σ_b , of the free group $F(\mathcal{C}_Z)$ as follows:

$$\forall c = w \cdot z \in \mathcal{C}_Z, \quad \sigma_b(c) = \begin{cases} w'c & \text{if } w = \bar{b}w'z \\ bc & \text{if } w \neq \bar{b}w'z \text{ and } bc \in \mathcal{C}_Z \\ c & \text{if } w \neq \bar{b}w'z \text{ and } bc \notin \mathcal{C}_Z. \end{cases}$$

The reader will easily check that σ_b is in fact an automorphism of $F(\mathcal{C}_Z)$ such that $\sigma_b^{-1} = \sigma_{\bar{b}}$. Then, arguing as in Section III.2, it is easy to prove that we can define a group morphism σ from $F(B, \mathcal{B}_B)$ into $\text{Aut}(F(\mathcal{C}_Z))$ as follows:

$$\sigma: F(B, \mathcal{B}_B) \rightarrow \text{Aut}(F(\mathcal{C}_Z)) \\ b \rightarrow \sigma_b.$$

Let us also consider the group morphism from $F(\mathcal{C}_Z)$ into $F(A, \mathcal{B})$ defined by

$$\forall c = w \cdot z \in \mathcal{C}_Z, \quad \alpha(c) = \pi(w) \cdot z \cdot \pi(w)^{-1}.$$

Let us finally introduce the mapping φ from the semidirect product of groups $F(\mathcal{C}_Z) \ltimes_{\sigma} F(B, \mathcal{B}_B)$ into $F(A, \mathcal{B})$ which is defined by

$$\forall c \in F(\mathcal{C}_Z), \forall b \in F(B, \mathcal{B}_B), \quad \varphi(c, b) = \alpha(c) \cdot b.$$

Then we have the following result which is the group version of Theorem III.3:

THEOREM IV.2 (Lazard's Partially Commutative Elimination). *Let A be an alphabet equipped with a partial commutation relation ϑ and let B be a subalphabet of A such that $Z = A - B$ is non-commutative. Then the mapping φ is an isomorphism from the free partially commutative group $F(A, \vartheta)$ into the following semidirect product of groups:*

$$F(A, \vartheta) \simeq F(\mathcal{C}_Z) \ltimes_{\sigma} F(B, \vartheta_B).$$

Proof. The reader can easily adapt the proof of Theorem III.3. ■

Remark. Now all the results of section III.2–3 can easily be expressed in terms of groups. For instance, we have also an internal version of Theorem IV.2 which says that the subgroup $\langle \mathcal{F}_Z \rangle$ of $F(A, \vartheta)$ generated by the family

$$\mathcal{F}_Z = \{ \pi(w) z \pi(w)^{-1}, w \in \text{Red}(M(B \cup \bar{B}, \Theta)), z \in Z, \text{TA}(\text{Alph}(wz)) = \{z\} \}$$

is free and that every element g of $F(A, \vartheta)$ can be written in a unique way as the product of an element of $\langle \mathcal{F}_Z \rangle$ by an element of $\langle B \rangle$ as follows:

$$g = c \cdot b \quad \text{with } c \in \langle \mathcal{F}_Z \rangle \text{ and } b \in \langle B \rangle.$$

In the same way, all the results of Section III.3 can easily be transposed to the group case. Here we shall only give the following result which can clearly be obtained from Theorem IV.2 by induction on $|A|$:

PROPOSITION IV.3. *Let A be a finite alphabet equipped with a partial commutation relation ϑ and let $\mathcal{F} = (A_k)_{k=1, n+1}$ be a total elimination for (A, ϑ) . Then the free partially commutative group $F(A, \vartheta)$ is isomorphic to a semidirect product of free groups of the form*

$$F(A, \vartheta) \simeq F(\mathcal{C}_{A_1 - A_2}(A_1)) \ltimes (F(\mathcal{C}_{A_2 - A_3}(A_2)) \ltimes (\dots \ltimes F(\mathcal{C}_{A_n - A_{n+1}}(A_n)) \dots)).$$

Notes. (1) It follows from the above result that a free partially commutative group on a finite alphabet A is always a semidirect product of free groups. This result gives in a certain sense an answer to a question of Diekert (cf. [Di.1]) since it says that the semidirect product is the good tool to decompose free partially commutative groups into free groups.

(2) Observe that the free partially commutative groups corresponding to the partial commutation relation λ_n introduced in the first example given at the end of Section II.3 is also a quotient of the braid group B_n or order n which has a presentation obtained by adding homogeneous relators of degree 3 to the relators λ_n (see [Ca, p. 716–10] or [CoMo]).

EXAMPLE. Let $n \geq 1$. Let us now consider an alphabet $A_n = \{a_i, i \in \llbracket 0, 2n \rrbracket\}$ equipped with a commutation relation ϑ_n defined by the commutation graph

$$\begin{array}{c} a_1 \text{ --- } a_2 \text{ --- } \dots \text{ --- } a_{2n} \\ a_0 \end{array}$$

According to our construction, let us first give the commutation graph of the extension Θ of ϑ to $A_n \cup \bar{A}_n$:

$$\begin{array}{c} \bar{a}_1 \text{ --- } \bar{a}_2 \text{ --- } \dots \text{ --- } \bar{a}_{2n} \\ \begin{array}{ccc} \diagdown & \diagup & \diagdown \\ a_1 & a_2 & \dots \\ \diagup & \diagdown & \diagup \end{array} \\ a_0 \text{ --- } \bar{a}_0 \end{array}$$

Now set $B_n = \{a_0\} \cup \{a_1, a_3, \dots, a_{2n-1}\}$. Then $Z = A_n - B_n$ is non-commutative and the code $\mathcal{C}_Z$ which is involved in Theorem IV.2 is given by the decomposition

$$\mathcal{C}_Z = \mathcal{C}_{\{a_{2n}\}}(A_n) \cup \bigcup_{k=1}^{n-1} \mathcal{C}_{\{a_{2k}\}}(A_n)$$

that we shall now make explicit. Let us now denote $\mathcal{B}_n = B_n \cup \bar{B}_n$. Then it is easy to see that we have

$$\mathcal{C}_{\{a_{2n}\}}(A_n) = \left(\mathcal{B}_n^* (\mathcal{B}_n - \{a_{2n-1}, \bar{a}_{2n-1}\}) \cap \left(\mathcal{B}_n^* - \bigcup_{b \in \mathcal{A}_n} \mathcal{B}_n^* b \bar{b} \mathcal{B}_n \right) \right) a_{2n}.$$

In the same way, we have for every $k \in \llbracket 1, n-1 \rrbracket$

$$\begin{aligned} \mathcal{C}_{\{a_{2k}\}}(A_n) = & \left(\mathcal{B}_n^* (\mathcal{B}_n - \{a_{2k-1}, \bar{a}_{2k-1}, a_{2k+1}, \bar{a}_{2k+1}\}) \right. \\ & \left. \cap \left(\mathcal{B}_n^* - \bigcup_{b \in \mathcal{A}_n} \mathcal{B}_n^* b \bar{b} \mathcal{B}_n \right) \right) a_{2k}. \end{aligned}$$

This gives a precise meaning to the following semidirect decomposition into two free groups which is given by Theorem IV.2:

$$F(A_n, \vartheta_n) \simeq F(\mathcal{C}_Z) \ltimes F(a_0, a_1, a_3, \dots, a_{2n-1}).$$

When $n=1$, the reader should compare this very simple decomposition to the negative result of Diekert (see [Di.2]) which said that $F(A_1, \vartheta_1)$ is not embeddable in any finite direct product of free groups or even in any weak product (see [Di.2]) of free groups.

(3) *Optimality of Lazard's Semidirect Decomposition*

As we showed in Section II.4, there exists a total elimination for (A, ϑ) with $\gamma(\vartheta)$ non-empty elements. Hence there exists an associated decomposition of $F(A, \vartheta)$ in a semidirect product of $\gamma(\vartheta)$ free groups that is given by Proposition IV.3. Observe also that in this decomposition, every letter of A lies in some of the free groups involved in it. We can show that Proposition IV.3 gives us again an optimal result in a certain sense:

THEOREM IV.4. *Let A be a finite alphabet equipped with a partial commutation relation ϑ and let us consider a decomposition of the group $F(A, \vartheta)$ into a semidirect product of free groups,*

$$F(A, \vartheta) \simeq F(\mathcal{C}_1) \ltimes (F(\mathcal{C}_2) \ltimes (\dots \ltimes F(\mathcal{C}_N) \dots)), \quad (*)$$

such that every letter of A lies in some $F(\mathcal{C}_i)$. Then the number N of free groups involved in $()$ is greater than or equal to the chromatic number $\gamma(\vartheta)$ of the commutation graph of ϑ .*

Proof. The reader can easily adapt the proof of Theorem II.8 to the group case. Indeed the two only arguments of this proof that are specific to the monoid case need only to be replaced by the following properties of a free group:

If x, y are two elements of a free group $F(X)$ which commute, then x and y are powers of a common element z of $F(X)$ (see [LySc, Proposition I.2.17]). Moreover one of these powers can be chosen to be positive.

..... If an element w of the free monoid X^* is equal to a positive power of an element x of the free group $F(X)$, then x is in X^* (see [LySc, Proposition I.2.15]).

This ends the proof of our result. ■

Remark. As in Section III.5, we conjecture that a semidirect decomposition of $F(A, \vartheta)$ into $\gamma(\vartheta)$ free groups is in fact optimal among all possible such semidirect decompositions of $F(A, \vartheta)$.

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